The Personal Agent Stack

A technical reference architecture for a consumer-scale personal AI agent, as of October 2026

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Research synthesised from six parallel research passes on 12 October 2026. Vendor-reported and unverified figures are labelled in the text. Prices and benchmark results change monthly; verify before committing money or making public claims.

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How to read this document

This is a technical reference, not a pitch. It describes what a state-of-the-art consumer personal agent is built from in October 2026, layer by layer: the models, the harness that drives them, the protocols they speak, the machines they act through, memory, connected accounts, proactivity, interfaces, payments, security, evaluation, observability, regulation, and unit economics. Each chapter ends with a concrete recommendation.

Evidence labels. Numbers come from six parallel research passes run on 12 October 2026. Where a figure is vendor-reported, aggregator-sourced, or seen only in a search excerpt, the text says so with *(vendor)*, *(aggregator)* or *(unverified)*. Benchmarks from different harnesses, effort settings or benchmark versions are not comparable; treat them as directional. Prices change monthly. Verify any figure before putting it in a pitch deck or contract.

Corrections to the 8 October paper are collected in Chapter 1 and flagged where they occur. The most important: a WhatsApp-first general assistant is *not permitted* for Indonesian (+62) users under Meta's current WhatsApp Business terms.

1 Executive summary

1.1 The one-paragraph version

In October 2026 every major lab ships the same product shape: a **per-user cloud computer** (sandbox plus browser), **connectors** into mail, calendar and apps, a **memory** that is curated in the background, **proactive** scheduled and event-triggered work, and **approval gates** in front of anything that spends money or speaks for the user. Meta Muse, OpenAI dots, Google Gemini Spark, Claude (Cowork, now cloud-default) and Manus all converge on it. The frontier model is no longer the differentiator: the cheap tier (Claude Haiku 5.5, GPT-6 Luna at \$0.10/\$0.50 per million tokens) now does computer use at a level the frontier reached a year ago, and both OpenAI and Anthropic will host your agent loop for "tokens only". What is still hard, and therefore defensible, is **reliability on long errands** (best binary completion on OSWorld 2.0 is still under 50%), **safe action** (prompt injection is contained by systems, not solved by models), **memory that users trust**, **access** (Google's restricted-scope audits, Meta's WhatsApp policy, Cloudflare's agent defaults, Amazon's terms), and **local rails** (payments, languages and services the US-first agents ignore). A consumer agent that wins in 2027 is a disciplined harness over rented models, with a trust layer good enough that people let it act.

1.2 Ten findings that should shape the build

1. **The cheap tier collapsed the cost curve.** Claude Haiku 5.5 (7 Oct 2026) costs \$0.10 input / \$0.50 output per million tokens for prompts under 100K and reports 72.4% on the OSWorld 2.1 offline subset *(vendor)*; GPT-6 Luna matches the price. A typical cached agent turn on a small model costs well under a tenth of a cent. Routing, caching and effort control, not model choice, decide gross margin.
2. **The harness is now a hosted product.** OpenAI's Agents API (public beta, 10 Sep 2026) and Anthropic's Claude Managed Agents (beta since 8 Apr 2026, with cron deployments and credential vaults since 9 Jun) give you compaction, tool search, sub-agents, sandboxes and scheduling without infrastructure. Build your own loop only if you need multi-model routing, custom memory, or data-plane control. For a consumer product with Indonesian data obligations, you probably do.
3. **MCP went stateless.** The 2026-07-28 revision removed the handshake and sessions, replaced server-initiated elicitation with multi-round-trip requests, deprecated dynamic client registration in favour of Client ID Metadata Documents, and formalised extensions (Tasks, MCP Apps, enterprise-managed auth). Tools, UI and auth for agents now have one standard path.
4. **Context engineering is the core skill.** Cache-stable prompts, progressive disclosure (skills and deferred tool search), programmatic tool calling, server-side compaction, a session log that lives outside the context window, and sub-agents used as context isolation. Manus calls cache-hit rate "the single most important metric".
5. **Multi-agent helps only where work is parallel and read-heavy.** Anthropic measured +90.2% on research at about 15× the tokens of chat; the Google/MIT scaling study found +80.9% on decomposable tasks but −39% to −70% on sequential planning. Keep writes single-threaded.
6. **Computer use is solved for short tasks, not long ones.** OSWorld-Verified is saturated (90.19% top entry vs a 72.36% human baseline). On OSWorld 2.0 (108 workflows, about 318 tool calls each) the best official binary completion is 44.33% (Claude Opus 5). Design errands as checkpointed, verifiable sub-tasks with a human hand-off.
7. **The web is formalising agent access.** Cloudflare stopped default-allowing "verified" bots (1 Jul 2026), blocks the Agent category on ad-carrying pages for new domains (from 15 Sep 2026), and is rolling out Web Bot Auth signatures and HTTP 402 payments. Disguising an agent as Chrome is now the riskiest strategy; signed identity, user-present sign-in and cooperative endpoints (WebMCP, UCP, ACP) are the durable ones.
8. **Memory converged, and its benchmarks broke.** Every serious system does write-time fact extraction, hybrid retrieval (vector + BM25 + entities + time), and an offline consolidation pass ("dreaming", sleep-time compute). LoCoMo's answer key is about 6.4% wrong; vendors report incompatible metrics.

Evaluate memory by task success with cost and latency.

9. **Payments layered out.** ACP (OpenAI/Stripe) and UCP (Google/Shopify) for checkout; AP2 (now at FIDO) for signed mandates; Visa Intelligent Commerce and Mastercard Agent Pay for scoped agent tokens; x402 and Stripe/Tempo MPP for machine micropayments. OpenAI pulled back in-chat Instant Checkout in March 2026: the model layer does not want to be the merchant. Indonesia is not yet in Visa's Agentic Ready programme; DOKU ships an MCP server for QRIS, virtual accounts and e-wallets.
10. **Regulation is arriving on known dates.** Indonesia's PDP implementing regulation (PP 33/2026) takes effect 16 Jan 2027. PP Tunas (in force since 28 Mar 2026) requires reliable age verification and bans ad profiling of children on high-risk platforms. The EU AI Act's Art. 50 transparency duties apply since 2 Aug 2026. California SB 243 regulates companion-like behaviour since 1 Jan 2026.

1.3 Corrections to the 8 October 2026 paper

What the earlier paper got wrong or is now out of date

1. WhatsApp-first is not allowed for a general assistant in Indonesia. Since 15 Jan 2026 Meta's WhatsApp Business Solution Terms bar "AI Providers" from offering general-purpose assistants as the primary function, except where law forces Meta to allow it (EEA including Italy, and Brazil with per-message charges). No Indonesian regulator has acted. Meta's own Muse runs natively inside WhatsApp. WhatsApp should be a notification and utility-template channel; the conversation lives in your own app and Telegram. Service replies also stopped being free in Indonesia on 1 Oct 2026 (first 1,000 per number per month free, then about Rp357 each before VAT).
2. The landscape moved. OpenAI's Atlas browser shut down on 9 Aug 2026 and ChatGPT Pulse was sunset on 17 Jun 2026 in favour of scheduled and event-triggered tasks. Google's Project Mariner was discontinued on 4 May 2026. Meta launched Muse on 8 Sep 2026 (US only). OpenAI launched dots on 29 Sep 2026.
3. Hosted harnesses change build-versus-buy. The earlier 16-week plan assumed you build the loop, sandbox wiring and scheduler. Much of that can now be rented; the plan in Chapter 15 reflects this.
4. Cost model. Small-tier inference prices fell sharply since the earlier estimate; proxies, voice and messaging fees now matter more than tokens for many users. Chapter 13 has a new model.
5. Browser profile advice needs a legal caveat. Persistent profiles and residential IPs still reduce 2FA friction, but large retailers increasingly block agents by terms of service (Amazon blocked Muse on 21 Sep 2026), even though the Ninth Circuit held on 4 Aug 2026 that a user-directed agent is the user under the US CFAA.

1.4 The converged architecture

Figure 1 is the reference architecture used throughout this document. The rest of the document fills in each box.

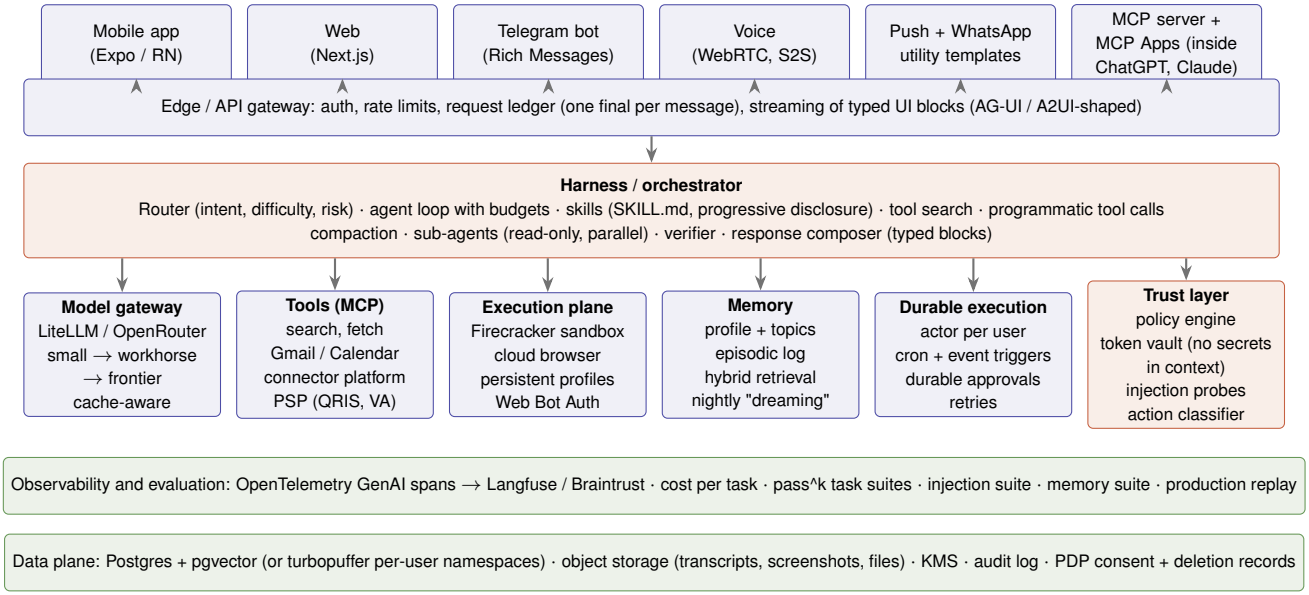


Figure 1. Reference architecture for a consumer personal agent (October 2026). Red boxes hold most of the defensible IP.

1.5 What is commodity and what is moat

Layer	Commodity in 2026?	Where advantage still comes from
Frontier and small models	Yes: four labs plus strong open weights, prices falling monthly	Routing, caching, effort control; fine-tuned small models on your own traces
Agent loop	Mostly: hosted by OpenAI and Anthropic, open SDKs everywhere	Skills library, recovery behaviour, verifiers, budgets, the request ledger
Sandboxes and browsers	Yes: $0.02–0.12 per browser-hour, $0.05–0.14 per vCPU-hour	Persistent identity, sign-in success rate, per-site playbooks, cooperative endpoints
Memory frameworks	Converging (Mem0, Zep, Letta, Cognee)	Data quality, a user-editable profile people trust, correct handling of sensitive and changing facts
Connectors	Partly (Composio, Arcade, Nango, Pipedream)	Passing Google CASA; depth in local services nobody else integrates
Interfaces	Standards exist (MCP Apps, A2UI, AG-UI)	Channel access, local language and voice quality, approval UX
Payments	Protocols exist, rollout uneven	Local rails (QRIS, VA, e-wallets) with scoped credentials and human approval
Trust and safety	No	System-level injection defence, audit trails, refund policy, regulatory readiness
Evaluation	No	Private task suites built from your own failures

1.6 Competitive bar, October 2026

Product	Shape	Price	Indonesia	Notes
Meta Muse (8 Sep 2026)	Agent on a per-user “Muse Secure VM”; app, web, Mac, iPad and inside WhatsApp	Free core, paid tiers	US only	Sentinel agent gates outbound actions; Link-by-Stripe one-time cards; Amazon blocked it on 21 Sep 2026

Product	Shape	Price	Indonesia	Notes
OpenAI dots (29 Sep 2026)	Always-on agents with their own cloud computer (GPT-6 Astra)	One dot in Pro ($100+) and Business Premium	Reportedly eligible (Pro from about IDR 1.89m/month) *(secondary)*	Proactive between conversations; Pulse sunset; Atlas retired
Google Gemini Spark (I/O, 19 May 2026)	24/7 cloud agent across Workspace, schedules and event triggers	AI Pro $19.99, Ultra $100/$200	Available in Bahasa Indonesia (Google ID blog, 29 Jul 2026)	Daily Brief; Personal Intelligence retrieval over Gmail and Photos
Claude (Cowork merged 16 Sep, cloud-default 6 Oct 2026)	General agent with cloud sandbox and built-in browser	Pro $20, Max $100/$200	Available	Claude in Chrome GA on paid plans (26 Aug 2026) with per-action classifier
Perplexity Comet + Computer	Free AI browser plus a credit-metered agent	Max $200	Available	Background assistants
Manus	Cloud-computer agent, Wide Research, scheduled tasks	$20–200	Available	Reportedly acquired by Meta *(unverified)*
Genspark	“Super Agent” workspace	About $25	Partly	Series B at $1.25B (Jun 2026)
Sahabat-AI (Indosat/GoTo)	Sovereign Indonesian LLM app	Free	Indonesia	Javanese, Sundanese, Batak, Balinese support; not a task agent

Gap to exploit. None of the global agents integrates QRIS, GoPay, OVO, DANA, ShopeePay, Tokopedia, Gojek/Grab, BPJS, KAI or Traveloka, and none is priced for a market anchored on ChatGPT Go. The largest single threat is Meta bringing Muse to WhatsApp in Indonesia, where it would own the default channel.

2 The model and inference layer

2.1 What changed in September and October 2026

Four things matter for an agent builder. The frontier moved (Claude Opus 5.5 on 22 Sep; GPT-6.1 Sol on 29 Sep; Gemini 4 Argon announced 30 Sep but restricted). The small tier became good enough to run browser and computer-use steps (Claude Haiku 5.5 on 7 Oct; GPT-6 Luna on 22 Sep). Open weights settled about one tier behind the frontier at a fraction of the price. And every provider made prompt caching explicit and cheap to read, which turns prompt layout into the main cost lever.

2.2 Frontier closed models

Prices are per million tokens at standard tier, short context, as listed on provider pricing pages on 12 Oct 2026. "AA" is the Artificial Analysis Intelligence Index v4.3.2; it is the only cross-vendor number here measured by one independent party.

Model	Released	Input / cached / output	Context	AA	Notable agent strengths
Claude Opus 5.5	22 Sep 2026	$4 / $0.20 / $20	1M / 128K out	58 (top)	Long-horizon knowledge work and computer use: 81.8% partial / 48.7% strict on OSWorld 2.0, 66.4% Terminal-Bench 4.0 at xhigh, 67.7% HLE with tools *(vendor)*
Claude Sonnet 5.5	28 Sep 2026	$2 / $0.10 / $10	1M / 128K	56	70.6% Terminal-Bench 4.0 *(vendor)*; cache read cut to $0.10 on 7 Oct
Claude Fable 5.1	1 Sep 2026	$10 / $0.25 / $50	1M / 128K	53	Highest-end Anthropic tier; adaptive thinking always on
GPT-6 Astra	4 Sep 2026	$10 / $1.00 / $50	1.05M / 128K	53	Leads some browser suites (VisualWebArena 78.5% in one third-party test, *unverified*); top Vending-Bench 2 score
GPT-6.1 Sol	29 Sep 2026	$2 / $0.10 / $10	1.05M / 128K	52	Best cost per task: about $0.72 per AA task vs $3.26 for Astra; 75.2% DeepSWE at high effort for about $0.65 per task *(vendor)*
Gemini 4 Argon	Announced 30 Sep 2026	Intro $2 / $0.10 / $10	Not officially stated	53	Not usable by startups yet (trusted cyber partners only)
Gemini 3.8 Flash	GA 2 Sep 2026	$0.75 / $0.075 / $3.75 until 31 Dec 2026, then double	1M / 65K	—	Cheap, fast, multimodal; 59.0% OSWorld 2.0 partial; weak on Terminal-Bench 4.0 (19.1%)
Grok 4.7	21 Sep 2026	$2 / $0.50 / $6	500K	46	Cheap output, live X data; AA hallucination rate 29%
Meta Muse Spark 1.3	2 Sep 2026	$1.25 / $0.15 / $4.25	1M	48	Closed API; launched US-only (check Indonesian availability)

Two practical notes. Claude 4.7 and later use a tokenizer that produces about 30% more tokens for the same text, so compare effective cost per task, not list price. And computer-use toolsets carry fixed overhead (about 4,500 tokens for Anthropic's computer-use toolset and 6,600 for browser-use), which caching should absorb.

2.3 The small tier: where most tokens should go

Model	Price (in / out)	Strengths	Use it for
Claude Haiku 5.5	$0.10 / $0.50 under 100K prompt; $0.50 / $2.50 above; cache read $0.01	72.4% OSWorld 2.1 offline subset (Luna: 48.9%); 39.2% Terminal-Bench 4.0; first Haiku with an effort setting *(vendor)*	Routing, browser steps, sub-agents, compaction, memory extraction
GPT-6 Luna	$0.10 / $0.50; cache read $0.01	1.05M context; effort from none to max	Classification, extraction, formatting, cheap chat
Gemini 3.5 Flash-Lite	$0.30 / $2.50	Multimodal, Search grounding	Vision triage, cheap grounding
DeepSeek V4.1-Flash (open, MIT)	$0.15 / $0.60 off-peak, $0.30 / $1.20 peak; cache hit $0.003	74.2 DeepSWE, 90.6 Terminal-Bench 2.1 *(excerpt)*	Batch and background work; self-hosting

Anthropic explicitly positions Haiku 5.5 as the executor under an Opus 5.5 planner. That is close to an off-the-shelf version of the planner/executor split recommended later in this chapter.

2.4 Open-weight models

Model	Params (total / active)	Licence	Fit
MiMo-V2.6-Pro (Xiaomi)	—	Open weights	Top open model on AA (46) at $0.435 / $0.87 hosted
GLM-5.3 (Z.ai)	~750B / 40B	Custom (GLM-5.2 was MIT)	Strong cheap workhorse (AA 45); JSON mode only, no strict schema; per-token price disputed
GLM-5.3-Flash	320B / 18B	MIT *(excerpt)*	Cheap multimodal extractor, ~$0.15 / $0.50
Kimi K3 (Moonshot)	2.8T / 104B	Custom	AA 44; 91.2% BrowseComp *(vendor)*; $3 / $15
Qwen3.8-27B	27.8B dense	Apache 2.0	Best self-hostable mid model: 84.3% OSWorld-Verified *(aggregator)*; ~18 GB quantised
Qwen3.5 0.8B–9B	Dense	Apache 2.0	Edge and router models
Gemma 4 (E2B, E4B, 12B, 26B-A4B, 31B)	—	Apache 2.0	On-device (E2B/E4B) and self-host; free on the Gemini API free tier
DeepSeek V4-Pro	1.6T / 49B	MIT	Text-only, $0.66 / $1.98 off-peak
MiniMax M3	~428B / 23B	Community licence	$0.30 / $1.20, cheap long context
Meta Muse Glimmer	30B	Apache 2.0	Laptop-class local agent model; ships with a DFlash speculative drafter
Mistral Small 4	119B / ~22B	Apache 2.0	Unified instruct/reasoning/coding with effort control

Licensing and residency caveats. Kimi K3, Qwen3.8-Max, GLM-5.3 and MiniMax M3 carry custom licences with commercial conditions (Qwen3.8-Max has a revenue share above $50M a year). Sending Indonesian users' personal data to China-hosted APIs creates data-transfer questions under the PDP Law; self-host or use US/EU/Singapore hosts (Bedrock, Together, Fireworks) for those models.

2.5 On-device models and runtimes

On-device inference is worth it for a narrow set of jobs: intent detection before a cloud call, PII redaction, notification triage, offline fallback and private quick replies. It is not a replacement for the cloud agent.

- **Apple Foundation Models, third generation** (WWDC, 8 Jun 2026): AFM 3 Core (3B dense) and AFM 3 Core Advanced (20B sparse, 1–4B active, weights streamed from flash; iPhone 17 Pro / Air and M3+ devices with 12 GB or more). The framework now offers guided generation (`@Generable`), tool calling, image input, a Private Cloud Compute model with 32K context, and pluggable third-party providers. On-device context is reportedly about 8K tokens *(unverified)*.
- **Gemini Nano 4** (production form of Gemma 4 E2B/E4B): up to 4× faster and 60% less battery than the previous Nano, but initially only seven phones. On Indonesia's mid-range Android base, ship a LiteRT-LM Gemma 4 E2B fallback (2.58 GB; about 52 tokens/s decode on a flagship GPU).
- **Runtimes:** llama.cpp (GGUF; MTP, EAGLE-3 and DFlash speculative decoding), MLX (Apple silicon), ExecuTorch, LiteRT-LM, Android AICore, Ollama and LM Studio for desktop.

2.6 Serving, hosting and gateways

Self-hosted engines. SGLang (v0.5.21; RadixAttention prefix tree and HiCache) is the better fit for agent loops that resend long prefixes; one H100 test showed parity with vLLM at a 50 ms per-token target and about 21% more capacity at 30 ms *(excerpt)*. vLLM (v0.31) wins on breadth of hardware and models; disaggregated prefill/decode is still experimental. TensorRT-LLM (stable v1.2.1) leads on NVFP4 and wide expert-parallel MoE at rack scale. NVIDIA Dynamo orchestrates disaggregated serving across all three.

Hosted inference. Cerebras serves gpt-oss-120b at about 1,700 tokens/s measured ($0.35 / $0.75); Groq about 480 tokens/s at $0.15 / $0.60. Together and Fireworks have the broadest open catalogues and fine-tuning. Bedrock and Vertex host Claude (regional endpoints cost about 10% more), which matters for data residency.

Gateways.

Gateway	Cost	Notes
LiteLLM (self-hosted)	Free (MIT core); enterprise by quote	Virtual keys, per-user budgets, fallbacks, 100+ providers; new Rust gateway claims 0.66 ms p99 overhead *(vendor)*
OpenRouter	5.5% fee on credit purchases; BYOK free to 1M requests a month	One key for 400+ models; its auto-router scored −24.7% vs the best single model in LLMRouterBench
Portkey	Free to 10K logs, then $49/month + $9 per 100K logs	Acquired by Palo Alto Networks (May 2026)
Vercel AI Gateway	0% markup	Simplest if deployed on Vercel; also proxies realtime audio
Cloudflare AI Gateway	Free core; 5% with unified billing	Caching, rate limits at the edge

2.7 Techniques that decide cost and latency

2.7.1 Prompt caching is the main lever

Provider	Cache write	Cache read	TTL and mechanics
OpenAI (GPT-5.6+)	1.25× input	0.1× (0.05× on GPT-6.1 Sol)	On by default; up to 4 explicit breakpoints; ≥30 min; `prewarm`; `configuration_update` changes effort without breaking the prefix
Anthropic	1.25× (5 min) / 2× (1 h)	0.1× standard; 0.05× Opus/Sonnet 5.5; Haiku 5.5 $0.01/M	Automatic or explicit `cache_control` breakpoints

Provider	Cache write	Cache read	TTL and mechanics
Google Gemini	Storage fee $0.50–4.50 per M tokens per hour	~0.1×	Implicit plus explicit caches
DeepSeek	None	2–3.3% of miss price	Automatic disk cache

Rules for an agent prompt, which apply to every provider:

1. Order the context as *stable policy prompt* → *tool definitions* → *skills index* → *user profile* → *conversation*, and only ever append.
2. Never put timestamps, per-request IDs or volatile user data in the prefix. Put the current time in the last user turn.
3. Do not edit the tool list mid-run; restrict callable tools with `allowed_tools` / `tool_choice` or deferred tool search instead.
4. Serialise JSON deterministically (sorted keys). MCP 2026-07-28 now asks servers to return tools in deterministic order for exactly this reason.
5. Use a per-user `prompt_cache_key` so one user's traffic cannot probe another's cache.
6. Compaction resets cache reuse; measure the total cost of a long session, not the per-turn cost.

OpenAI reports one multi-turn agent deployment above a 90% cache-hit rate; Manus reports input-to-output ratios around 100:1 in agent loops. At those ratios, a cached read at 0.05–0.1× is the difference between a viable and an unviable free tier.

2.7.2 Batch, flex and off-peak

OpenAI, Anthropic and Gemini batch and flex tiers are all 50% off. DeepSeek's off-peak window covers Jakarta evenings and nights (peak is 08:00–11:00 and 13:00–17:00 WIB on weekdays). Use batch for memory consolidation, nightly briefs, proactive scans, evaluation runs and embedding backfills.

2.7.3 Reasoning effort is a second routing dial

Every major family now exposes effort (OpenAI `reasoning.effort`, Anthropic `effort` with adaptive thinking, Gemini `thinking_level`, xAI and DeepSeek `reasoning_effort`). More effort is not monotonically better: GPT-6.1 Sol at high effort beats its own max setting on DeepSWE (75.2% vs 71.9%) at under half the cost *(excerpt)*. Route on (model, effort) pairs, and let the harness raise effort on retry rather than by default.

2.7.4 Routing: hand-designed beats black-box

RouteLLM (2024) cut cost over 85% at 95% of GPT-4 quality on MT-Bench. But LLMRouterBench (ACL Findings 2026; 400K+ instances, 33 models) found the best routers save at most about 32% at equal quality, many learned routers do no better than simple baselines, and a popular commercial auto-router came out at −24.7%. MTRouter (ACL 2026) shows multi-turn, per-step routing for agents can cut cost 58.7% on ScienceWorld. The practical conclusion: a small classifier that routes on task type and risk, plus escalation on failure or low confidence, applied per step rather than per conversation.

2.7.5 Speculative decoding and structured output

DFlash (block-diffusion drafter) reports up to 6× lossless speedup and about 2.4× over EAGLE-3 under greedy decoding; NVIDIA reports 5.3–5.8× on Gemma 4 31B at concurrency 1. Speedups shrink as concurrency rises, so this mainly helps latency on self-hosted, low-QPS models (voice, on-device). Strict JSON-schema output is supported by OpenAI, Gemini, xAI, DeepSeek, Qwen3.8 and Kimi K3; GLM-5.3 offers JSON mode only. On OpenAI the output schema is part of the cached prefix, so do not vary schemas between calls.

2.8 Recommended tiering

Tier	Job	Primary	Alternates
T0 on-device	Intent, PII scrub, offline replies	Apple Foundation Models; Gemini Nano 4 / Gemma 4 E2B	Qwen3.5-4B
T1 router, classifier, extractor	Every turn; JSON from mail and pages	Claude Haiku 5.5 or GPT-6 Luna	DeepSeek V4.1-Flash off-peak; self-hosted Qwen3.8-27B or Gemma 4 12B
T2 workhorse	Most tool-using turns, browsing, bookings	GPT-6.1 Sol or Claude Sonnet 5.5	Gemini 3.8 Flash (until price doubles 1 Jan 2027), GLM-5.3
T3 frontier escalation	Long-horizon, high-stakes, hard computer use	Claude Opus 5.5	GPT-6 Astra; Gemini 4 Argon when public
Batch / background	Consolidation, digests, evals, backfills	Any of the above on batch (−50%)	DeepSeek off-peak

Worked cost per turn (my arithmetic, list prices, cache writes ignored). A 30K-token context, 90% cached, with 1K output on GPT-6.1 Sol costs about 27K × \$0.10/M + 3K × \$2/M + 1K × \$10/M ≈ **\$0.019**. A 20K context with 500 output on Haiku 5.5 costs about **\$0.0006**. A 60K-context Opus 5.5 step with 2K output costs about **\$0.075**. One unnecessary escalation to the frontier costs as much as 120 small-model turns; the router is the margin.

3 The agent harness

The harness is everything between the model and the product: the loop, tool dispatch, context management, sub-agents, budgets, recovery, verification and presentation. In 2025 it was a library you assembled; in 2026 both frontier labs sell it as a hosted service. The question is no longer "how do I build an agent loop" but "which parts of the loop do I need to own".

3.1 Hosted harnesses

OpenAI Agents API (public beta, 10 Sep 2026). The Codex harness offered as a managed service. A session is created with a model, MCP tools, optional multi-agent settings (`max_concurrent_subagents`), credential vault IDs and an environment (OpenAI-hosted sandbox, self-hosted, or partners such as Cloudflare, Daytona, E2B, Modal, Vercel, Blaxel, Runloop, DigitalOcean, Oracle). It includes automatic compaction, tool search, programmatic tool calling, sub-agents and skills via capability directories. There is no fee beyond tokens and tools. On 29 Sep it gained computer use in an OpenAI-hosted browser with per-origin approval. Caution: OpenAI's own guide notes that origin approval "does not enforce confirmation before individual actions", so once a store is approved the agent can check out. Gate spending at the credential, not at the origin.

Claude Managed Agents (public beta, 8 Apr 2026). A hosted harness plus sandbox plus durable session log, with APIs for agents, environments, sessions and events. Sessions run for hours and survive disconnects. Since launch it added self-hosted sandboxes and MCP tunnels (19 May), "dreaming" scheduled memory curation, outcomes and multi-agent orchestration (19 May), and scheduled deployments plus vaults that inject credentials at the network boundary so the sandbox only sees placeholders (9 Jun). Runtime is billed at $0.08 per session-hour plus tokens. A "dynamic workflows" feature (9 Oct) is reported but unverified.

Anthropic's engineering lesson from building it is worth adopting regardless of vendor: *decouple the brain from the hands*. The session log lives outside the context window and is sliced into context on demand; harness assumptions go stale as models improve, so keep them replaceable.

3.2 Self-hosted SDKs

SDK	Version (Oct 2026)	Strengths	Best for
OpenAI Agents SDK	Python ~0.23; JS 0.18 (~2.3M weekly npm downloads)	Agents, handoffs, guardrails, sessions, tracing, realtime; Sandbox Agents (0.14, Apr 2026) with snapshot/resume across Docker, E2B, Modal, Daytona, Cloudflare, Vercel	OpenAI-first teams; Temporal integration GA
Claude Agent SDK	Python 0.2.165 (8 Oct); TS 0.3.29x	The Claude Code loop as a library: sub-agents, hooks, permission callbacks, MCP, skills, compaction; `verbatimPrompts` for untrusted text	Claude-first; strongest coding and computer harness. Isolate `settingSources` in multi-tenant backends
Google ADK 2.x	Python 2.10–2.11; Go, TS, Kotlin (Android, on-device via LiteRT-LM)	Graph runtime mixing deterministic and LLM nodes; built-in human-in-the-loop; skill lifecycle	Gemini, Android and on-device
LangGraph + Deep Agents	LangGraph 1.2; Deep Agents 0.7 (−65% base input tokens)	Model-neutral; checkpointers; skills whose tools are added after the cached prefix	Deep customisation
Mastra	core 1.73	TypeScript; MCP 2026-07-28 server; Temporal adapter; per-request model selection	TS full-stack products

SDK	Version (Oct 2026)	Strengths	Best for
Pydantic AI 2.x	2.55	Typed Python; unified cross-provider caching; durable execution on seven engines (Temporal, DBOS, Restate, Prefect…)	Typed Python backends
Vercel AI SDK 7	7.0 (25 Jun 2026); 16M+ weekly downloads	`ToolLoopAgent`, durable `WorkflowAgent`, `HarnessAgent` wrapping Claude Code / Codex / Deep Agents in a sandbox, tool approval policies with signed replay, MCP Apps renderer, OTel	Next.js consumer UIs
Microsoft Agent Framework	1.0 GA (Apr 2026)	Merges Semantic Kernel and AutoGen; graph workflows; A2A	Enterprise / .NET
CrewAI	1.15.x	Role-based flows	Enterprise automations; had chained CVEs in Mar 2026

Recommendation. For a consumer agent with Indonesian data obligations and multi-model routing, own the loop on an SDK (TypeScript: AI SDK 7 or Mastra; Python: Pydantic AI or the OpenAI Agents SDK), and rent sandboxes and browsers underneath. Use a hosted harness (Agents API or Managed Agents) for prototypes, for internal tools, and as a fallback executor for long coding-style jobs.

3.3 The agent loop, made concrete

The loop from the earlier paper is still right; what changed is what wraps it.

```
on message(m):
  req   = ledger.open(m)                    # every message owed exactly one final
  plan  = router.classify(m, profile)       # intent, difficulty tier, risk, skills
  ctx   = assemble(policy_prefix,           # cache-stable
                   tools.deferred_index,    # names only; full schemas on demand
                   skills.index,            # ~100 tokens each
                   memory.recall(m, plan),  # 2-7K tokens, cited
                   session.tail())          # log lives outside the window
  for step in budget(plan.tier):
      out = model(plan.model, plan.effort, ctx)
      if out.final:  return compose(out) -> typed blocks -> ledger.close(req)
      calls = policy.check(out.tool_calls, provenance(ctx))   # trust layer
      results = run_parallel(calls)          # durable, idempotent, timed out
      ctx.append(summarise_if_large(results))   # full output to disk, path in ctx
      if loop_detector.same_step_twice(): escalate(model or effort) or ask_user()
      if ctx.near_limit(): ctx = compact(ctx)
  return compose_partial(ctx)               # say what is done and what is not
```

Four additions matter in 2026. **Tool results are stored outside the context** with only a summary and a path kept in it (reversible compression). **A loop detector** escalates the model or effort, or asks the user, instead of letting the agent flail. **Policy checks see provenance**, so an action derived from untrusted content can be blocked or sent for approval. **Partial completion is a first-class result**: telling the user exactly what finished is better than a silent failure.

3.4 Context engineering: eight rules

These rules come from Anthropic's context-engineering and long-running-harness posts (2025–2026), OpenAI's Codex and harness-engineering posts, and Manus's production lessons.

1. **Design around the KV cache.** Stable prefix, append-only history, deterministic serialisation, no tool-list mutation mid-run (mask instead). Cache-hit rate is the metric.

2. **Progressive disclosure.** Skills expose about 100 tokens of metadata and load their body only when triggered. Tools are deferred and found through tool search (Anthropic `defer_loading`, OpenAI `tool_search` with namespaces); past roughly 30 tools, deferral improves both accuracy and cost.
3. **Programmatic tool calling.** Let the model write code that calls tools in a sandbox and returns only filtered results. Anthropic reports +11% on BrowseComp and DeepSearchQA with −24% input tokens.
4. **Server-side compaction as a backstop.** OpenAI's `context_management` compaction and `/responses/compact`; Anthropic's compaction beta and tool-result clearing. For some models, a context reset with a structured hand-off artifact beats compaction.
5. **The filesystem is memory.** Keep full tool outputs on disk, URLs and paths in context, a `todo.md` the agent re-reads to keep its goal in view, and error traces in context so it learns within the task.
6. **Session is not context.** A durable event log outside the model is the source of truth; the context window is a view over it.
7. **Sub-agents are for context isolation.** Give each a clean window, an explicit objective, an output format and boundaries; return condensed results or file references.
8. **Long tasks need an initializer.** An initializer agent writes a feature or task list (all "failing"), a progress file and a git history; workers each complete one item per session and update progress; a separate evaluator checks.

3.5 Skills as the product's procedural memory

A skill is a folder with a `SKILL.md` file (name of 64 characters or fewer, description of 1,024 or fewer, optional scripts, references and assets). It became an open standard on 18 Dec 2025 (agentskills.io) and is supported by about 40 products, including Claude, ChatGPT/Codex, Gemini CLI, Copilot, Cursor and Manus *(third-party tally)*. For a consumer agent, skills are where hard-won procedure lives: how to cancel a subscription on a given site, what to confirm before booking, how to recover when a sign-in fails. They are versioned in git, A/B testable, writable by non-engineers, and loaded only when relevant.

Treat skills as code with a supply chain. OWASP's Agentic Skills Top 10 lists malicious skills, weak isolation and update drift as risks. Only load first-party or reviewed skills into a context that also holds user secrets or payment tools.

3.6 Multi-agent: when it helps

Evidence	Finding
Anthropic Research system (2025)	Opus lead + Sonnet sub-agents beat a single Opus by 90.2% on an internal research eval; agents use ∼4× chat tokens and multi-agent ∼15×; token use explained 80% of BrowseComp variance
Google Research / DeepMind / MIT scaling study (Nature Machine Intelligence, Jul 2026)	Centralised multi-agent +80.9% on decomposable tasks; every multi-agent variant −39% to −70% on sequential planning; once a single agent exceeds ∼45%, coordination returns diminish; independent agents amplify errors 17.2×; practical ceiling of 3–4 agents
Cognition (Apr 2026)	Works best when writes stay single-threaded and extra agents contribute intelligence (review, research), not actions
"The Illusion of Multi-Agent Advantage" and "The Interaction Tax" (2026)	Equal-budget comparisons often show no gain; diversity, not count, drives improvement

Design rule. One main agent owns all writes, purchases and messages. Spawn up to three to five parallel read-only sub-agents for breadth (flights, hotels, research across sources). Add a clean-context verifier before irreversible actions. Budget tokens by tier.

3.7 Durable execution

Personal agents wait: for a reminder date, for a user's approval, for a site to come back, for a multi-hour browser task. Anything that waits must survive deploys and crashes.

Engine	Model	Agent integrations	When to pick
Temporal	Replay-based workflows; cluster or cloud	OpenAI Agents SDK integration GA (23 Mar 2026); ADK, AI SDK, Mastra, Pydantic AI	Most mature; heavier operations
Inngest	Step functions; managed or self-host	AgentKit; Durable Endpoints; sessions	Serverless TypeScript
Restate	Journaled handlers, virtual objects; single binary	AI SDK, OpenAI Agents, ADK, Pydantic AI; 1.7 flow control	Lightweight actor-per-user
DBOS	Library; checkpoints in Postgres	OpenAI Agents, Pydantic AI, AI SDK 7	Cheapest if you already run Postgres
Trigger.dev	Checkpoint-resume TS tasks	`chat.agent` GA (2 Jul 2026)	Durable chat turns
Cloudflare Agents SDK	Durable Object per agent (SQLite, alarms, hibernation) + Workflows V2	Scheduled tasks, background sub-agents, MCP 2026-07-28	Edge, cheapest idle, actor per user
Hosted harnesses	Managed Agents cron; Agents API sessions	Built in	Zero infrastructure

One third-party estimate puts a 200K-runs/month agent workload at $33–411/month on these engines, under 1% of token spend *(unverified)*. Durability is cheap; losing a user's half-finished booking is not.

Pattern. One actor per user (a Durable Object or Restate virtual object) holds the user's queue, schedules and in-flight tasks. Every model call and tool call is a checkpointed step with an idempotency key. Human approvals are durable waits with a timeout and a reminder. Secrets are injected at the egress proxy, never stored in workflow state.

3.8 Response composition

Separate reasoning from presentation. The model emits typed blocks (message, table, summary, order card, links, draft, question) that each client renders natively. This enforces brevity, keeps the app, Telegram and voice consistent, makes outputs testable, and lets the same result render as an A2UI surface, an MCP App or a WhatsApp template. A request ledger (every user message owed exactly one final answer, with an explicit state machine) prevents duplicate, dropped or out-of-order replies when several sub-agents finish at different times.

4 Protocols and standards

By late 2026 the agent protocol stack has a clear layering, most of it governed by the Linux Foundation's Agentic AI Foundation (AAIF, formed 9 Dec 2025):

Layer	Standard	Status (Oct 2026)	Role in a consumer agent
Tools and data	MCP 2026-07-28	Current; ~0.5B SDK downloads a month	How the agent reaches every tool, internal or third-party
In-chat UI from tools	MCP Apps (`io.modelcontextprotocol/ui`)	Stable since 26 Jan 2026	Rich cards inside ChatGPT, Claude, Copilot, VS Code, and your own client
Agent to frontend	AG-UI 1.0	Frozen 18 Sep 2026	Event stream between your agent backend and your apps
Declarative native UI	A2UI	v0.9.1 production, v1.0 candidate	JSON component streams rendered by native components (Flutter, React, Compose)
Agent to agent	A2A v1.0	Stable since 12 Mar 2026; in AAIF since 27 Aug 2026	Delegating to third-party vendor agents (low priority today)
Procedural knowledge	Agent Skills (SKILL.md)	Open standard since 18 Dec 2025	Your skills library
Repo instructions	AGENTS.md	60K+ projects	Your own coding agents, not the product
Website tools for agents	WebMCP	Spreading (Shopify checkout 28 Sep 2026, PayPal, ChatGPT desktop browser)	Cooperative sites expose tools instead of pixels
Agent identity on the web	Web Bot Auth (RFC 9421 signatures)	Cloudflare, AWS WAF, Akamai and others	Signed requests instead of spoofed browsers
Commerce	ACP, UCP, AP2, x402, MPP	See Chapter 9	Checkout, mandates and payments

4.1 MCP revision 2026-07-28

The largest revision since launch. What changed and why it matters:

- **Stateless core.** No `initialize` handshake and no `Mcp-Session-Id`; every request carries protocol version and client capabilities in `_meta`. Servers can run behind round-robin load balancers or in a plain serverless function. Cross-call state is carried by server-minted handles passed as tool arguments.
- **Multi round-trip requests (MRTR).** Server-initiated `elicitation/create`, `sampling/createMessage` and `roots/list` are replaced: the server returns `resultType: "input_required"` with `inputRequests`, and the client retries with `inputResponses`. This is how confirmations and forms now work, and it maps cleanly onto a durable approval wait.
- **Routing and caching.** Required `Mcp-Method` and `Mcp-Name` HTTP headers let gateways and WAFs route without parsing JSON. List and read results carry `ttlMs` and `cacheScope`. Servers should return tools in deterministic order for prompt-cache stability.
- **Subscriptions.** A single long-lived `subscriptions/listen` stream; SSE resumability removed.
- **Auth hardening.** RFC 9207 issuer validation; credentials bound to the issuing authorization server; Dynamic Client Registration deprecated in favour of Client ID Metadata Documents (CIMD); Protected Resource Metadata (RFC 9728) mandatory; audience-bound tokens (RFC 8707). Enterprise-Managed Authorization is an extension. The TypeScript SDK added DPoP.
- **Extensions framework.** Reverse-DNS IDs. First extensions: Tasks (long-running jobs with `tasks/get` and `tasks/update`, contributed by AWS), MCP Apps, and Enterprise-Managed Authorization.

- **Deprecated** with at least a 12-month window: Roots, Sampling, Logging, legacy HTTP+SSE, DCR.

The official MCP Registry (`registry.modelcontextprotocol.io`) is still in preview with an API frozen at v0.1; about 17,000 servers were published by July 2026 *(third-party tally)*. Do not let the agent install arbitrary registry servers into a context that holds user data.

4.2 MCP Apps and generative UI

A tool declares `_meta.ui.resourceUri` pointing at a `ui://` HTML resource. The host renders it in a sandboxed iframe, talks to it only through `postMessage` JSON-RPC, and enforces a server-declared Content Security Policy. Hosts include ChatGPT, Claude, VS Code, Microsoft 365 Copilot, Goose, Postman and Cursor. OpenAI's Apps SDK is now MCP Apps plus ChatGPT-only extras (checkout, modals, file pickers), and the app directory became a "Plugin" directory on 9 Jul 2026.

Google's A2UI takes the other approach: the agent streams a declarative JSON description (`createSurface`, `updateComponents`, `updateDataModel`) and the client renders it with its own native components from an advertised catalogue. Google's guidance is to combine A2UI for native, safe surfaces with MCP Apps for custom iframes. AG-UI 1.0 is the transport between agent backend and frontend and integrates with LangGraph, CrewAI, Mastra, Pydantic AI, ADK, Microsoft Agent Framework and the Claude Agent SDK.

The consumer baseline moved: ChatGPT's GPT-6 "Intelligent UI" (7 Oct 2026) streams interactive charts, forms and maps to every user, and Claude added Dashboards and Motion (8 Oct 2026). Text-only answers now feel dated.

Recommendation. Define your own typed block catalogue (message, table, order card, approval, form, map, chart), shaped like A2UI so it renders natively in React Native and on the web. Publish your agent's capabilities as an MCP server with MCP Apps UI so it also runs inside ChatGPT, Claude and Copilot, which is a distribution channel you do not have to build.

4.3 A2A, AGENTS.md and WebMCP

A2A v1.0 brought signed Agent Cards, JSON-RPC, gRPC and REST bindings, multi-tenancy and version negotiation; 150+ organisations back it. For a consumer agent it matters later, when airlines, banks or marketplaces expose their own agents. AGENTS.md is a coding-agent convention: useful for your engineering workflow, not for the product. WebMCP is the one to watch for execution: when a site exposes checkout or search as tools to in-browser agents, the agent stops guessing at pixels. Shopify's checkout and PayPal's JS SDK already support it.

5 The execution plane: acting in the world

5.1 The action ladder

The single most important execution-plane decision is the order in which the agent tries ways of acting. Cheaper, more reliable, more legitimate paths come first:

1. **Official API or MCP tool** (Gmail, Calendar, a payment provider's MCP server such as DOKU's, a merchant's MCP server). Deterministic, auditable, cheapest.
2. **Cooperative agent endpoints**: WebMCP tools exposed by the page, UCP or ACP checkout endpoints. Structured, and the site has opted in.
3. **DOM and accessibility-tree automation** (Playwright MCP, Stagehand, Browser Use): reads the page structure, no vision model needed for most steps.
4. **Pixel-level computer use** (Claude, Gemini or OpenAI computer-use models): the universal fallback, slowest and most expensive per step.

Every skill should record which rung works for its site. Most consumer errands on large Indonesian platforms will land on rungs 3 and 4 for years, which is exactly why per-site playbooks and persistent identity are defensible.

5.2 Computer-use models and what the benchmarks mean

Benchmark	What it measures	State, October 2026
OSWorld-Verified	369 short desktop tasks, ∼30 tool calls each; human 72.36%	Saturated: top official entry 90.19% (Intelligence-Indeed Agent, 25 Jul 2026); Claude Fable 5 85.96%; Claude Opus 5 83.39%
OSWorld 2.0 / 2.1	108 long workflows across 31 self-hosted sites and pro apps; ∼318 tool calls; ∼1.6 h human median	Best official: Claude Opus 5 (max) 44.33% binary / 77.67% partial. Vendor-reported: Opus 5.5 48.7% / 81.8%; Fable 5.1 41.7% / 77.9%; Gemini 3.8 Flash 59.0% partial
Online-Mind2Web	300 live tasks on 136 sites	Microsoft Fara1.5-27B 72.3%; Gemini 2.5 Computer Use 69.0%
WebArena / VisualWebArena	Reproducible self-hosted web apps	GPT-6 Astra 78.5% vs Opus 5.5 70.1% on 177 VWA tasks *(unverified)*
BrowseComp	Hard-to-find facts on the live web	Near-saturated tie at ∼90–92% (GPT-5.6 Sol, Kimi K3, GPT-6 Astra, Opus 5)
ERPBench (Sep 2026)	Enterprise software	Open models with strong OSWorld scores collapse (UI-TARS 42.5 → 9%)

How to read these numbers: partial scores far exceed binary completion on the same run; benchmark releases are not comparable (Opus 5 went from 31.43% to 44.33% binary between OSWorld 2.0 releases with no model change); effort and step budget move results by 2×; and one OSWorld 2.0 task cost about $72 at about 482 tool calls for Opus 4.8. For a consumer agent this means **long errands still succeed end-to-end only about half the time and are expensive**. Design for checkpoints, verifiable sub-goals, and a graceful hand-back to the user.

Model choice by job. Anthropic (Opus 5.5, Fable 5.1, and now Haiku 5.5 for cheap steps) leads long-horizon desktop and GUI work. OpenAI GPT-6 Astra leads some browser suites. Gemini 3.8 Flash is the cheap fast option with built-in computer use billed as normal tokens (the old `gemini-2.5-computer-use-preview` is legacy). Open models (Qwen-CUA at 86–87.6% OSWorld-Verified, Holo3-35B-A3B at 82.6%, Fara1.5, EvoCUA, OpenCUA) are close on short benchmarks and worth self-hosting for cheap, privacy-sensitive steps, but degrade on real enterprise and long-tail sites.

Consumer reference points. Claude in Chrome became generally available on all paid plans on 26 Aug 2026 and now acts autonomously with a per-action safety classifier instead of per-action approval, while still asking before financial sites. OpenAI's Atlas browser stopped working on 9 Aug 2026; agentic browsing moved into the ChatGPT desktop app and Codex. Google discontinued Project Mariner on 4 May 2026 in favour of Chrome "auto browse" (US preview). OpenAI reported zero-intervention success on 8–16 hour tasks rising from 10% (Jan 2026) to 35% (Jul 2026) *(unverified)*.

5.3 Cloud browsers

Provider	Price per browser-hour	Proxy	Max session	Notable
Browser Use Cloud	$0.02 (1-min minimum)	$5/GB residential	4 h	Hardened Chromium fork, auto CAPTCHA, profile and cookie sync, 1Password + TOTP; OSS library (~118K stars)
Kernel	~$0.06 headless, $0.48 headful, per second	Managed	72 h	Standby after 5 s idle stops billing; fastest session creation in its own test (37 ms)
Anchor	$0.05 + $0.01 per start	$8/GB	3 h	Managed auth identities; BYOC on enterprise
Steel	$0.10 ($0.08 on Scale)	$10 / $6 per GB	up to 24 h	Apache-2.0 self-hostable runtime
Hyperbrowser	$0.10 per second-metered hour	$10/GB	12 h	Hosted agent steps at $0.02
Browserbase	$20/month incl. 100 h then $0.12/h; $99 incl. 500 h then $0.10/h	$10–12/GB	6 h	Stagehand v3 (CDP-native, `verified: true` agent identity, WebMCP), Search and Fetch APIs; advanced stealth on Scale only

Stealth success varies by who runs the test: Browser Use's own 71-site test put itself at 81% and Browserbase basic stealth at 42%; a third-party BrowserBench (296 anti-bot tasks) put Browser Use at 84.8% and Browserbase at 70.3%. Every vendor wins its own benchmark. Run a proof of concept on *your* target sites (Tokopedia, Shopee, Traveloka, tiket.com, KAI, PLN, bank portals) before choosing.

Cost reality. Browser-hours are cheap; residential proxy bandwidth at $5–12 per GB and model tokens dominate. A media-heavy checkout flow can pull 20–50 MB; block images and video where the task allows.

Local control. Playwright MCP (v0.0.83, accessibility-snapshot driven, `--extension` mode attaches to the user's real Chrome profile) and Chrome DevTools MCP (v1.10.1, ~58 tools, `--autoConnect` to a signed-in Chrome) let a desktop companion act inside the user's own browser session. That is the most reliable path for sites that must see the user's real device and session, at the cost of needing the user's machine online.

5.4 Identity, sign-in and the changing rules of the web

Persistent identity. A new browser fingerprint looks like a new device to Google, banks and marketplaces. One persistent profile per user per site (cookies, local storage, fingerprint) plus a stable egress IP region turns "a 2FA prompt every task" into "one prompt per month". Store profiles encrypted, scoped to one user, never shared.

Signed agents instead of disguise. Web Bot Auth signs requests with HTTP Message Signatures (RFC 9421), with the operator's Ed25519 keys published at `/.well-known/http-message-signatures-directory`. Since 1 Jul 2026 Cloudflare merged signed agents and verified bots and stopped default-allowing them: access now follows each site's Search / Agent / Training category settings. For domains onboarded from 15 Sep 2026, Training and Agent are blocked by default on ad-carrying pages. Cloudflare's Monetization Gateway (beta, 30

Sep 2026) lets sites charge agents per request via HTTP 402 settled over x402.

The legal picture. In *Amazon v. Perplexity*, the Ninth Circuit (4 Aug 2026; rehearing denied 10 Sep) vacated an injunction and held that a user-directed agent is the user accessing the site under the US Computer Fraud and Abuse Act. Contract and terms-of-service claims remain open, and Amazon blocked Meta's Muse on 21 Sep 2026 citing its conditions of use. Indonesian law has no equivalent ruling; marketplace terms apply.

Implication. Disguising an agent as an ordinary Chrome user is now the riskiest path, technically and legally. Sign requests, sign in with the user present (secure forms for codes and approvals, secrets straight to the vault), prefer cooperative endpoints, and pursue partnerships with the handful of local platforms that matter most.

5.5 Code sandboxes

The sandbox is where the agent computes, renders documents, runs data analysis and executes generated code. Rule from the earlier paper, still the biggest accuracy lever for STEM: never compute in the model's head when a sandbox can compute.

Platform	Isolation	CPU price	Memory price	Billing	Session cap
E2B	Firecracker microVM	$0.0504/vCPU-h	$0.0162/GiB-h	Wall-clock while running; paused is free	1 h hobby / 24 h pro
Daytona	Containers (VM and Windows classes available)	$0.0504/vCPU-h	$0.0162/GiB-h	Wall-clock	Configurable
Modal Sandboxes	gVisor	~$0.071/vCPU-h	$0.024/GiB-h	max(request, actual)	24 h
Vercel Sandbox	Firecracker microVM	$0.128/vCPU-h active only	$0.0212/GB-h	Active CPU	45 min / 24 h
Cloudflare Sandbox SDK	Containers on Workers	$0.072/vCPU-h active	$0.009/GiB-h	Active CPU; disk resets on sleep	Sleeps after 10 min idle
Fly.io Sprites	Firecracker	$0.07/CPU-h	$0.044/GB-h	Active only, persistent 100 GB disk	Persistent

Under a 100-concurrent burst, median cold starts were 0.67 s (Vercel), 0.88 s (Modal), 1.61 s (E2B) and 5.06 s (Cloudflare); Daytona started fastest but succeeded only 37% of the time under burst. For idle-heavy agent loops (10 minutes alive, 5% CPU), cost per 1,000 runs ranged from about $11 to $40, or about $2 with E2B auto-pause. **Billing model matters more than headline rate.**

Isolation. Firecracker microVMs give each sandbox its own guest kernel (strongest for untrusted, LLM-written code); gVisor intercepts syscalls in user space (lighter, weaker); plain containers are weakest. Default egress to deny, allow-list per task, and inject credentials at the egress proxy (Cloudflare outbound handlers, Vercel brokered credentials, E2B request transforms) so a prompt-injected agent cannot read a token it never had. Gotchas: E2B's `onTimeout` defaults to kill (set pause); E2B allow-rules beat deny-rules while Vercel's deny beats allow.

5.6 Recommended execution plane

- **Routing:** small model (Haiku 5.5 or Gemini 3.8 Flash) for routine browser steps; escalate to Opus 5.5 or GPT-6 Astra for long or high-stakes flows; budget for failure and hand-back.
- **Browser:** one managed provider chosen by a POC on Indonesian targets (Browser Use Cloud on cost and stealth, Browserbase on tooling and identity, Kernel on latency and idle billing), with persistent per-user profiles, Indonesian residential egress, Web Bot Auth signing where accepted, and a desktop-companion path through Playwright `--extension` for sites that need the real session.

- **Sandbox:** Firecracker class (E2B with auto-pause, or Vercel for idle-heavy loops), deny-by-default egress, credentials at the proxy.
- **Verification:** after any action, a cheap verifier checks independent evidence (confirmation email, account page, order list) before the user is told “done”.

6 Memory and personalisation

Memory is what turns a tool into something people rely on, and it is where agents most visibly fail: forgetting a stated preference, recalling a stale one, mistaking a gift purchase for the user's own taste, or reciting private details at the wrong moment.

6.1 The converged architecture

Every serious memory system in 2026 has the same four parts:

1. **Write-time extraction.** An LLM turns each conversation (or email, or task) into atomic facts with a timestamp, source pointer, confidence and type.
2. **A hybrid store.** Vectors for meaning, BM25 for exact terms, an entity table for people, places and organisations with aliases, and time validity on each fact.
3. **Multi-signal retrieval.** Dense + keyword + entity match + recency, fused (usually reciprocal rank fusion) and reranked, under a token budget.
4. **Offline consolidation.** A background pass dedupes, merges, expires past-dated facts, rewrites the profile summary, and detects contradictions. Letta calls it sleep-time compute, OpenAI "dreaming", Cognee "memify".

Temporal awareness is now table stakes: "You're going to Singapore in July" must become "You went to Singapore in July 2026" without user intervention.

6.2 Memory systems compared

System	Architecture	Reported results	Notes
Mem0	ADD-only single-pass extraction (no update/delete at write; recency resolved at retrieval); fused semantic + BM25 + entity scoring; built-in entity linking (replaced Neo4j option); temporal reasoning and decay (May 2026)	LoCoMo 92.5, LongMemEval 94.4, BEAM-1M 64.1, BEAM-10M 48.6; ∼7K tokens/query; p50 ∼1 s *(vendor; managed platform, OSS "directionally similar")*	Also authored DolphinBench
Zep / Graphiti	Bi-temporal knowledge graph: episodes → entities and fact edges with valid/invalid times → communities; new facts invalidate old edges, enabling "as-of" queries	LongMemEval 90.2%; retrieval p50/p95 104/162 ms; ∼4.4K context tokens *(vendor)*	Graphiti is open source (Neo4j/FalkorDB)
Letta (MemGPT)	Core memory blocks in context + archival/recall memory; self-editing via tools; sleep-time agents; git-backed MemFS (Feb 2026)	Sleep-time compute: ∼5× less test-time compute at equal accuracy (paper)	Argues a plain filesystem agent is competitive
Cognee 1.0	Extract → cognify → load; graph + vector + BM25 with RRF; four verbs: remember, recall, forget, improve	No primary benchmark number	Zero-infra embedded stack; on-device Rust core
LangMem	Semantic, episodic and procedural (prompt-optimisation) memory; hot-path tools or background manager	78.05 LoCoMo in a third-party run	LangGraph store
Supermemory	Graph where facts update, connect and expire; built-in profiles	"97%" is LongMemEval-S *Recall@20*, not QA accuracy	Most expensive and least accurate on DolphinBench (Mem0-authored)

System	Architecture	Reported results	Notes
MemOS	Parametric, activation (KV) and plaintext memory as governed "MemCubes"	LoCoMo 88.83, LongMemEval 89.20, BEAM-10M 56.75 *(vendor)*	Research-heavy
A-Mem (NeurIPS 2025)	Zettelkasten notes with links and memory evolution	LoCoMo across six base models	Good design ideas for linking

6.3 Do not trust memory benchmarks at face value

- **LoCoMo** is near-saturated and about 6.4% of its answer key (99 of 1,540) is wrong; one judge configuration accepted about 63% of deliberately wrong answers.
- **Metrics differ**: retrieval recall@k, QA accuracy and task success are reported interchangeably.
- **Newer, better tests**: BEAM (1M and 10M tokens), LongMemEval-V2 (agentic contexts), and DolphinBench (Sep 2026), which requires the agent to *use* memory to complete 600 verified tool tasks over ∼500K-token histories and to report cost and latency. Its best result was 70.67% (Hermes + GPT-5.6-Luna + Mem0); Claude Code with built-in memory scored 26.33% at $1,132. Rankings flip across harnesses and models.

Evaluate on your own action-based tasks with cost and latency. A memory system that lifts LoCoMo but slows every turn by a second is a regression for a consumer product.

6.4 How the big three do it

ChatGPT. Saved memories (2024) and chat-history reference evolved into "Dreaming V3" (4 Jun 2026): a single background-synthesised, time-aware memory state, reviewable and editable as a memory summary page, cheap enough (∼5× less compute) to serve free users. Answers now show which memories shaped them, with correct, delete and "not relevant" controls. Deleting a chat does not delete derived memories; turning off chat-history reference schedules them for deletion within 30 days.

Claude. Two features. Chat search is RAG over past conversations, invoked as a visible tool call with citations. Memory (2026 design) saves individual topics *during* the chat rather than a post-hoc summary, with per-project spaces, per-chat off switch and incognito. Sensitive topics (health, religion, politics) are excluded by default and opt-in; government IDs, financial account numbers, criminal history and immigration status are never stored. Developers get a client-side, file-based memory tool (`/memories`) and, in Managed Agents, versioned memory stores mounted into the sandbox.

Gemini Personal Intelligence. Instead of a static profile, Gemini retrieves on demand from live Gmail, Photos, YouTube and Workspace, using tool calls and Gemini embeddings, and "context packs" the relevant pieces because personal data exceeds the 1M-token window by orders of magnitude. Simple queries take a fast non-personalised path.

Google's Personal Intelligence whitepaper lists failure modes that make an excellent design checklist:

Personalisation failure modes to design and test against

- **Tunnel vision**: over-personalising every answer around one known interest.
- **Someone else's preference**: gifts, shared accounts and family purchases read as the user's own taste.
- **Timeline mix-ups**: future plans treated as past events and vice versa.
- **Misread relationships**: wrong inference about who a person is to the user.
- **Missed life changes**: a divorce, a death or a move not reflected; stale facts surfaced painfully.
- **Confirmation is not completion**: a booking email treated as a completed action while a later cancellation or return is ignored.

- **Overlooked corrections**: the user corrected a fact and the old one returns.

6.5 Memory principles for a consumer agent

1. **Write the core, not the particulars.** Store decisions and preferences with their conditions ("window seat on long-haul only"), not search results, prices or one-off details.
2. **Never more certain than the user.** Mark inferred facts as inferred, with what they rest on; confirm before acting on them. "Usually" never becomes "always".
3. **Recency wins conflicts, by rewriting.** Contradictions are resolved by updating the fact (with a short note of what it replaced), not by appending a second line.
4. **Sensitive by default is off.** Health, religion, sexuality, finances and minors' data need explicit opt-in; never proactively infer them.
5. **Visible and deletable.** A human-legible profile ("what you know about me") that users can correct, plus validated true deletion: a trust feature and a PDP Law requirement.
6. **Recall is retrieval plus judgement.** Hybrid search returns candidates; the model decides relevance and never recites memory unprompted. Cite memory sources in answers when they shaped the result.
7. **Conditions travel with facts.** A preference has a place, people and occasion. Save them, and ask before applying a preference from one context to another.

6.6 Data model

```
memory_fact(
  id, user_id, subject_entity_id, predicate, object, conditions jsonb,
  stated_or_inferred, confidence, sensitivity, source_ref,
  valid_from, valid_to, superseded_by, created_at, embedding vector, tsv tsvector)
entity(id, user_id, kind[person|place|org|project], canonical_name, aliases[], relation_to_user)
profile(user_id, typed fields: name, timezone, addresses, dietary, accessibility, standing_rules)
episode(id, user_id, day, summary, embedding)          -- daily log
approval_rule(user_id, action_class, scope, decision, stated_at)
```

6.7 Retrieval stack

Embeddings. The MTEB leaderboard has fragmented (English, Multilingual v2, RTEB retrieval). Strong options: Qwen3-Embedding-8B (70.58 multilingual mean-task, Apache 2.0, 32K context) or Qwen3-Embedding-4B for self-hosting; Voyage 4 family ($0.02–0.12 per M tokens; one shared embedding space across sizes, so documents can be embedded with the large model and queries with the lite one); Gemini Embedding. OpenAI's `text-embedding-3` models remain current; a "text-embedding-4" is not confirmed. Test Bahasa Indonesia and code-switched queries explicitly.

Rerankers. Voyage rerank-3 / rerank-3-lite (30 Sep 2026; $0.05 / $0.02 per M tokens) and Cohere Rerank 4 (32K context, 100+ languages). Rerank roughly 150 candidates down to 20.

Contextual retrieval. Anthropic's measurement remains the reference: prepending a 50–100 token context to each chunk before embedding and BM25 indexing cut top-20 retrieval failure from 5.7% to 2.9%, and to 1.9% with a reranker (−67%), at about $1 per million document tokens with caching.

Vector stores. pgvector 0.8.6 (stay on the latest patch; iterative index scans fix filtered-query under-return) is right up to roughly tens of millions of vectors and keeps memory transactional with app data. turbopuffer (object-storage native, namespace per tenant, BM25 plus vectors; minimum cut to $16/month) suits one-namespace-per-user mail and document indexes at scale. Qdrant (OSS, filter-aware HNSW, server-side hybrid) and LanceDB (embedded, columnar) are good alternatives.

6.8 Recommended memory design

- **Write path:** per-turn extraction on the small model into an append-only log; nightly consolidation on the batch API that dedupes, expires, merges, rewrites the profile summary and queues contradictions as questions for the user.
- **Store:** Postgres (pgvector + tsvector) with entity and alias tables; move per-user mail and document indexes to turbopuffer namespaces if scale demands.
- **Read path:** query rewrite with profile → hybrid (dense + BM25 + entity boost + recency decay) → rerank → 2–7K token budget, plus a memory-search tool for just-in-time recall instead of stuffing context.
- **User layer:** an editable profile and topic list, memory citations in answers, a per-chat memory-off switch, export and verified deletion.
- **Evaluate:** a multi-session memory suite (set, change, delete, sensitive, conflicting) scored on task success, false recall, deletion completeness, cost and latency.

7 Connected context and proactivity

7.1 Connected accounts: the policy wall

The technology for reading mail and calendars is easy. The policy is not, and it should shape the product roadmap.

Google scope tiers.

Tier	Example scopes	Requirement
Non-sensitive	`gmail.labels`, `drive.file`	None beyond standard consent
Sensitive	`gmail.send`, most Calendar scopes	OAuth app verification
Restricted	`gmail.readonly`, `gmail.compose`, `gmail.modify`, full Gmail, most Drive	Verification **plus** an annual third-party CASA security assessment whenever data is stored on or passes through your servers

CASA (Cloud Application Security Assessment, based on OWASP ASVS) is tiered by risk and must be redone every 12 months. Google charges nothing; assessors set prices. Google's own documentation cites **2–3 weeks for Tier 2 and 4–6 weeks for Tier 3, with fees of roughly $500–4,500**; one lab quotes $5,400 for a Tier 3 manual test. Unverified apps are capped at 100 test users. Whether on-device-only Gmail processing escapes CASA is officially unresolved.

Design implications. Launch with scopes that avoid CASA (Calendar, `gmail.send`, `gmail.labels`, `drive.file`) and features that need them (calendar briefings, sending drafts the user approved, organising labels). Budget CASA as a recurring line item before inbox-reading features. Consider on-device extraction for mail on mobile, sending only derived facts to the server, but get a written answer from Google before relying on it.

Microsoft Graph. Delegated `Mail.Read`, `Calendars.Read` and `Files.Read.All` work for personal accounts; in work tenants with Microsoft's recommended consent policy, users cannot self-consent, so admin consent is needed. Publisher verification needs a Microsoft Partner Network ID.

Connector platforms.

Platform	Positioning	Best for
Composio	Agent-native tool calling, managed OAuth, triggers; ~1,000 toolkits	Breadth quickly. A May 2026 security incident is reported only by a competitor *(unverified)*
Arcade.dev	MCP runtime with per-user, scope-pinned, just-in-time auth; tokens never reach the model	Safety-first tool execution
Pipedream Connect	3,000+ apps via managed proxy	Long tail
Nango	Open-source auth + proxy + data syncs + webhooks + MCP auth	Syncing data for retrieval, not just tool calls
Merge Agent Handler	Unified APIs and governed MCP tool packs with PII scanning	Enterprise

None of these removes your own CASA obligation when you use your own Google OAuth client for restricted scopes. Most published comparisons are written by competitors.

7.2 Proactivity: from feeds to triggers

The industry tried a morning feed and moved on. ChatGPT Pulse (Sep 2025) generated 5–10 cards each morning; it was sunset on 17 Jun 2026 "as proactive updates move into scheduled tasks". Its replacement has three primitives: **scheduled tasks** (up to hourly on top plans), **event-triggered tasks** on Gmail, Slack and GitHub (up to 30 per hour), and **monitoring tasks** that "notify users only when there is something worth reporting". Unattended tasks auto-pause after inactivity, and side-effecting actions pause for approval.

Google's Gemini Daily Brief (I/O, May 2026) splits "Top of mind" (timely, actionable items with one-tap actions) from "Looking ahead" (long-term goals), and Gemini Spark adds event and conditional triggers. Meta's Muse states the principle most directly: *a proactive message has to be genuinely helpful and worth the interruption, and users can turn it off, dial it down or dial it up.*

7.3 Eight rules for proactive messages

1. **Interrupt budget.** A high bar per push and a user-adjustable dial; default to a digest over a push.
2. **Triggers over polling.** Monitor and speak only on meaningful change.
3. **Actionable cards.** Each item carries a one-tap next action (reply, schedule, cancel, remind) and why it surfaced.
4. **Feedback becomes rules.** Dismiss, "less of this" and "stop reminding me about X" are standing rules that nothing else loosens.
5. **Approval gates.** Proactive work prepares; it does not act on money or messages without a tap.
6. **Freshness disclosure.** Pre-computed briefs go stale; show "as of" times.
7. **Sensitive restraint.** Never proactively infer health, relationships or life events; handle bereavement and divorce with care and without follow-up nags.
8. **Auto-pause on disengagement.** If a user stops opening cards, slow down before they churn.

7.4 Implementation

- **Watchers** run on the batch tier: new mail and calendar events are triaged (needs user / waiting on others / informational / ignore) by the small model; only "needs user" with a concrete offer becomes a card.
- **Schedules** live in the per-user actor (Chapter 3) with idempotent firing and catch-up rules after downtime.
- **Delivery** goes to push first; WhatsApp utility templates only for high-value, time-sensitive items (each costs about $0.025 in Indonesia).
- **Metrics:** card open rate, action rate, dismiss rate, "less of this" rate and uninstalls within seven days of a card. Optimise action rate per card, not cards sent.

8 Interfaces: channels, apps and voice

8.1 Messaging channels

8.1.1 WhatsApp: what is and is not allowed

Date	Event
15 Oct 2025	Meta adds an "AI Providers" clause to the WhatsApp Business Solution Terms: LLM and general-purpose-assistant providers may not use the Business API where AI is the primary (not incidental) function, "as determined by Meta in its sole discretion"
Dec 2025 – Jan 2026	Italy's AGCM orders interim suspension (Meta exempts +39); Brazil's CADE orders suspension (+55 exempted)
15 Jan 2026	Ban takes effect; OpenAI, Perplexity and Microsoft shut down their WhatsApp bots
4 Mar 2026	Meta readmits rivals in Europe for €0.049–0.1323 per non-template message
9 Jun 2026	European Commission imposes interim measures: free access on pre-ban terms until the investigation ends (at the latest 2029); ChatGPT returns across the EEA on 13 Jul
8 Sep 2026	Meta's own Muse launches, reachable "directly in WhatsApp" (US)
1 Oct 2026	Indonesian rate card changes: service replies no longer free beyond 1,000 per number per month

For Indonesia: Meta's current policy page allows general-purpose assistants from AI providers "only where Meta is legally required to permit this use case". No Indonesian regulator (including KPPU) has acted. **A general assistant on the WhatsApp Business API for +62 users is not permitted**, and unofficial WhatsApp Web automation violates the terms and risks number bans.

Allowed patterns: (a) a business using AI to serve its own customers; (b) utility and authentication templates (notifications, OTPs, "your task is done, tap to review" with a deep link); (c) a narrowly scoped vertical service where AI is ancillary, which is a grey zone needing legal review.

Indonesian WhatsApp prices from 1 Oct 2026 (per delivered message, before 11% VAT and BSP fees; from consistent secondary sources, Meta's own table not read directly):

Category	Price
Marketing	Rp586.33 ($0.0411)
Utility	Rp356.65 ($0.025)
Authentication	Rp356.65 ($0.025)
Service (free-form replies within 24 h)	First 1,000 per number per month free, then Rp356.65

Even where allowed, a chatty agent is now expensive on WhatsApp: 1,000 users sending ten replies a day is about 300,000 service messages a month, roughly Rp107 million (about $7,500) before VAT.

8.1.2 Telegram is the open alternative

Telegram has moved aggressively towards AI bots in 2026: streaming drafts for all bots (Bot API 9.5), managed bots that create bots (9.6), Guest Mode to tag an AI bot in any chat and bot-to-bot chats (10.0, 8 May 2026), and Rich Messages with tables, maps, maths and "thinking" blocks plus rich-message streaming, inline buttons and ephemeral overlays (10.1–10.3). The API is free (about 30 messages/s global, 1 per second per chat), payments run through Telegram Stars. Reach in Indonesia is meaningful but well below WhatsApp's *(share not verified)*.

8.1.3 Other channels

RCS Business Messaging launched in Indonesia through Telkomsel with Google and Infobip (Dec 2025), with mandatory brand verification and strong campaign results (Blibli reported 2.5× conversion); good for verified Android notifications, unproven as a conversational surface, pricing unpublished. LINE matters in Thailand, Taiwan and Japan, not Indonesia. iMessage has no open bot API for general assistants.

Channel strategy. Own app first (where approvals, cards, payments and memory controls live), Telegram second as the no-install surface, WhatsApp utility templates for nudges and OTPs, push for everything else. Publish an MCP server with MCP Apps so the agent is also reachable inside ChatGPT, Claude and Copilot.

8.2 Client stack

Mobile. Expo SDK 54–55 with React Native 0.81–0.83 and React 19. Use `expo/fetch` for streaming (React Native's global `fetch` has no stream body). The Vercel AI SDK `useChat` works in Expo. Render a typed block catalogue natively; `@json-render/react-native` streams JSON-spec UIs into native components. Push (Expo Notifications, FCM, APNs) carries "agent finished" and "needs your approval".

Web. Next.js with the Vercel AI SDK (v6 transport, v7 agents): typed UI message streams, tool-call UI parts, approval components. Fluid compute for long streams.

Approval UX is the product. Every competitor has converged on: a plan the user can see, an order card with merchant, items, fees, total and payment method before any charge, a one-tap approve button tied to that exact total, a visible audit trail, and one-time virtual cards rather than raw card access (Muse uses Link-by-Stripe one-time cards). Design these screens first.

8.3 Voice

8.3.1 Architecture choice

- **Speech-to-speech (S2S):** one model listens and speaks; lowest latency, natural barge-in and prosody. You cannot inspect a reply before it is spoken and you are locked to one vendor.
- **Cascaded (STT → LLM → TTS):** every stage swappable, text you can log and filter before speech; more latency per hand-off.
- **Hybrid "thinker/talker" (the 2026 pattern):** a full-duplex voice model handles conversation while a separate reasoning model does the work. OpenAI GPT-Live-1 (10 Sep 2026; $0.05/min voice layer plus backend tokens) and Gemini 3.8 Live Extended Thinking (asynchronous function calls while it keeps talking) both work this way.

Rule: the voice layer never commits an irreversible action on its own. Pay, send and book go through the text agent with an explicit confirmation card or spoken read-back.

8.3.2 Prices

Option	Price	Notes
Gemini 3.8 Live (GA 15 Sep 2026)	~$0.005/min listening, ~$0.018/min speaking	97+ languages including Indonesian; async function calls; Google claims #1 on Artificial Analysis S2S board
OpenAI gpt-realtime-2.1	~$0.019/min listening, ~$0.077/min speaking	Cached audio input is 80× cheaper than uncached; wire caching correctly
gpt-realtime-2.1-mini	~$0.006 / ~$0.024 per min	
OpenAI GPT-Live-1	$0.05/min + reasoning model	Full duplex; sideband monitor socket
STT: Soniox stt-rt-v5	$0.12/hour	Vendor-reported 1.25% Indonesian WER; code-switching in one stream

Option	Price	Notes
STT: Gemini 3.5 Transcribe	Not stated	Intra-sentence code-switching (useful for Indonesian–English mixing); 1,000-term custom vocabulary
STT: Deepgram Nova-3	~$0.0077/min	One reviewer measured 14.2% WER on Bahasa Indonesia
TTS: Cartesia Sonic 3.5/3.6	~$0.03/min	Indonesian supported; independent median latency ~269–440 ms
TTS: ElevenLabs Flash v2.5	~$0.05 per 1K characters	Indonesian supported; independent median ~185 ms
Framework: LiveKit Agents	From $0.01/min overage; SIP $0.003–0.004/min	Apache-2.0; semantic turn detection; native SIP
Framework: Pipecat Cloud	$0.01 per active minute	Regions US, EU, India only

Estimated all-in cost per conversation minute (50/50 talk ratio, excluding telephony): **Gemini 3.8 Live** ≈ **$0.02–0.03**, gpt-realtime-2.1-mini ≈ $0.03, a cascaded stack ≈ $0.03–0.05.

Latency targets: P50 under ~400 ms and P95 under ~800 ms to first audio (humans leave about 208 ms between turns). Check which Singapore or Jakarta points of presence your providers have; a US round trip adds 60–200 ms.

Indonesian quality is unproven. No independent Bahasa Indonesia S2S or TTS benchmark exists; every quality claim above is a vendor's or a single reviewer's. Test on real local speech: code-switching, Javanese and Sundanese loanwords, regional accents, numbers and currency ("Rp1,5 juta"), and noisy mobile networks. Voice notes (asynchronous STT → agent → TTS) may matter more than live calls in Indonesia, where people favour WhatsApp voice notes over phone calls.

Telephony. Indonesian +62 numbers require Komdigi KYC (KTP or passport and local address for individuals; an Indonesian entity and NPWP for companies), 3–10 business days. Twilio lists a local number at $23/month and SIP termination at $0.0266/min. Call recordings are personal data under the PDP Law.

8.4 OS assistants and devices

Android AppFunctions lets apps expose functions as on-device tools, but calling them requires a privileged permission that effectively only Gemini holds (private preview as of May 2026). A third-party agent can expose its own app to Gemini but cannot call other apps. **iOS 27** Siri acts through App Intents; reports of third-party model extensions are unverified as shipped. With roughly 90% Android share in Indonesia *(unverified)*, Gemini is the default OS-level competitor.

Glasses and devices. Meta Ray-Ban Display ($799, US, UK, Canada, and parts of the EU from 13 Oct 2026) is not sold in Southeast Asia; Muse is coming to Meta glasses and a 5G handheld, Muse Charm. For Indonesia this is a 2027+ consideration; design voice-first flows (phone plus earbuds) that can port later.

9 Agentic commerce and payments

9.1 The protocol stack

Layer	Protocol	Owner	Status (Oct 2026)	What it does
Checkout and discovery	ACP (Agentic Commerce Protocol)	OpenAI + Stripe	Live; spec "beta", latest 17 Apr 2026	Cart, feed, orders; Shared Payment Token scoped to merchant and cart total; OpenAI is not merchant of record
Checkout and discovery	UCP (Universal Commerce Protocol)	Google + Shopify, with Amazon, Meta, Microsoft, Salesforce, Stripe on its council	Live in the US; v2026-08-25 adds 3DS2, split pay, instalments, loyalty	Discovery at `/.well-known/ucp`; uses AP2 mandates and RFC 9421 signatures
Authorisation	AP2 (Agent Payments Protocol)	Google → FIDO Alliance (28 Apr 2026)	v0.2, pre-1.0	Signed checkout and payment mandates as verifiable credentials; v0.2 adds human-not-present payments
Network credentials	Visa Intelligent Commerce / Trusted Agent Protocol	Visa (TAP built with Cloudflare on Web Bot Auth)	Live in Europe (2 Jul 2026); Agentic Ready in 10 APAC markets	Agent tokens; one integration accepting TAP, MPP, ACP and UCP
Network credentials	Mastercard Agent Pay	Mastercard	Live authenticated agentic transactions in Singapore and Malaysia (Apr 2026)	Agentic tokens bound to agent and merchant scope; Verifiable Intent donated to FIDO
Identity interop	Know-Your-Agent	Visa + Mastercard + Ant International	Announced Sep 2026	Aligning TAP, Verifiable Intent and Ant's Agentic Mobile Protocol
Machine payments	x402	x402 Foundation (Linux Foundation), 50+ members	Live on-chain	HTTP 402 stablecoin payments; ~75M transactions / ~$24M per 30 days (Jul 2026), inflated by wash trading; no chargebacks
Machine payments	MPP (Machine Payments Protocol)	Stripe + Tempo	Live since 18 Mar 2026	Stablecoins plus cards via SPTs; Link agent wallet issues one-time cards with per-purchase approval

9.2 Lesson from OpenAI Instant Checkout

OpenAI launched in-chat Instant Checkout on ACP in September 2025 (Etsy first). On 24 Mar 2026 it said the first version "did not offer the level of flexibility that we aspire to provide" and moved purchases to merchant-owned checkout and merchant apps, while extending ACP to product discovery with large US retailers. In-chat payment survives inside merchant apps (Instacart, Walmart). **The model layer did not want to own checkout**: fraud, returns, taxes and disputes belong to the merchant. A consumer agent should drive the merchant's checkout (or a merchant app) using a scoped, single-use credential, not become a merchant of record.

9.3 Shopify and PayPal

Shopify's Agentic Storefronts push catalogues into ChatGPT, Copilot and Google AI Mode; Google's direct checkout is on by default for eligible US stores; Shopify checkout exposes WebMCP tools to browser agents since 28 Sep 2026. PayPal's Agent Ready pays through ACP delegated tokens and UCP's Google Pay handler, Store Sync distributes catalogues to Copilot, Meta and Perplexity, and its JS SDK adds WebMCP.

9.4 Southeast Asia and Indonesia

- **Visa Agentic Ready** went live on 30 Apr 2026 in Australia, Hong Kong, Japan, Malaysia, New Zealand, Singapore, South Korea, Taiwan, Thailand and Vietnam. **Indonesia is not listed.** DBS was the first APAC issuer to pilot Visa Intelligent Commerce.
- **Mastercard** has authenticated agentic transactions live in Singapore and Malaysia, with an ASEAN rollout planned.
- **Indonesian rails** are e-wallet and virtual-account heavy (GoPay, OVO, DANA, ShopeePay; QRIS as the mandatory unified QR; BI-FAST for real-time transfers). Foreign payment providers need a PJP licence under PBI 10/2025 (effective 31 Mar 2026); most products integrate through Xendit, Midtrans or DOKU.
- **QRIS merchant fee is 0% up to IDR 100,000** for all merchant categories from 1 Oct 2026 (micro merchants up to IDR 500,000), alongside the new Kartu Kredit Indonesia domestic credit card paid by QRIS.
- **DOKU MCP Server** is the first Indonesian gateway with an MCP server: 30+ tools covering payment links, QRIS, virtual accounts, cards, e-wallets, paylater, convenience-store codes and transaction lookup. DOKU says it contributes to AP2.
- No agent-payment launches found from GoPay, OVO, DANA or Xendit *(searches only; recheck)*. Bank Indonesia has publicly framed AI governance and payments as priorities (6 Oct 2026) but has no agent-payment rule.

9.5 Recommended payment design

1. **Never give the agent raw card numbers.** Use scoped, single-use credentials (Stripe Shared Payment Tokens or Link one-time cards, Visa and Mastercard agent tokens where available), with per-merchant and per-amount caps.
2. **Gate spending at the credential, not the origin.** Approving a site is not approving a purchase. Each purchase gets an order card with the exact total; the tap releases a credential valid for that merchant and amount only. Keep an AP2-style signed mandate log.
3. **Total changed? Ask again.** If the checkout total moves beyond tax and reasonable fees, stop and present a fresh card.
4. **Local rails through a licensed PSP.** For QRIS, virtual accounts and e-wallets, generate the payment through a licensed provider's API or MCP server (DOKU, Xendit, Midtrans) and let the user complete it in their own wallet app. The agent prepares; the user pays.
5. **Micropayments are separate.** x402 or MPP for paying APIs and data on the agent's own budget, never for consumer retail (no chargebacks).
6. **Minors.** Restrict or disable purchases for users under 18; Indonesia's PP Tunas and most platform terms push in this direction.
7. **Verification and evidence.** Store the confirmation number, a screenshot and the merchant's confirmation email in the audit log before telling the user "done".

10 Trust and security

10.1 Threat model: the lethal trifecta

Simon Willison's framing is the right starting point: an agent that combines **access to private data**, **exposure to untrusted content** and **the ability to communicate externally** can be tricked into exfiltrating that data. A consumer personal agent holds all three by design: it reads the user's mail, browses arbitrary web pages and sends messages or makes purchases. The goal is not to avoid the trifecta (the product would be useless) but to make sure no single execution path holds all three legs without a gate.

OWASP's two lists give the vocabulary:

OWASP Top 10 for Agentic Applications 2026	How it shows up in a personal agent
ASI01 Agent goal hijack	An email or web page instructs the agent to forward mail or buy something
ASI02 Tool misuse and exploitation	Legitimate tool used with attacker-chosen arguments (wrong recipient, wrong amount)
ASI03 Identity and privilege abuse	Agent's broad OAuth scopes used beyond the task
ASI04 Agentic supply chain	Malicious skill, MCP server or connector
ASI05 Unexpected code execution	Generated code escapes the sandbox or reaches the network
ASI06 Memory and context poisoning	Injected "fact" written into long-term memory
ASI07 Insecure inter-agent communication	Sub-agent output trusted as instructions
ASI08 Cascading failures	One wrong step compounds across a long errand
ASI09 Human-agent trust exploitation	Approval fatigue; user taps "approve" on a manipulated card
ASI10 Rogue agents	Agent pursues a goal beyond what was asked

The LLM Top 10 (2025) adds sensitive information disclosure, system prompt leakage, vector and embedding weaknesses, and unbounded consumption (cost attacks).

10.2 What the labs report

Source	Setting	Attack success rate
Anthropic, Claude in Chrome GA (Aug 2026)	New red-team eval, no safeguards	Opus 4.5 17.6%; Opus 5 3.8%
Same	With injection probes + action classifier	0% for Sonnet 5, Opus 5, Mythos 5; 0.3% for Fable 5 (low severity)
Anthropic (Nov 2025)	Opus 4.5 vs adaptive best-of-N attacker	~1%
Anthropic system card (via press)	129 held-out environments × 10 tries, no safeguards	Sonnet 4.6 50.7% → Opus 4.8 31.5%; Opus 4.8 with safeguards 0.5%
OpenAI GPT-Red (Jul 2026)	Self-play RL red-teamer vs GPT-5.6 Sol, direct injections	0.05%; "fake chain-of-thought" attacks fell from >95% to <10%
Google DeepMind CaMeL (SaTML 2026)	AgentDojo with provable security	77% tasks solved vs 84% undefended; 0 successful attacks in one 949-attack run
NIST CAISI (via CSA note)	Custom AgentDojo red-team attacks on production models	81% *(secondary)*

Two lessons. First, **the 0% figures are system-level**: the raw model behind an API ships without those probes and classifiers, so you must build the equivalent. Second, OpenAI says plainly that prompt injection "is unlikely to ever be fully solved". Plan for containment.

10.3 Defence patterns

Beurer-Kellner et al. (2025) define six patterns with provable properties; a consumer agent uses several at once:

- **Action-Selector**: the model picks from a fixed set of actions and never sees tool output. Good for simple commands.
- **Plan-Then-Execute**: the plan is fixed before untrusted content is read; content can fill parameters but not change the plan.
- **LLM Map-Reduce**: untrusted items processed by isolated sub-calls whose outputs are constrained (for example, booleans or enums).
- **Dual LLM**: a privileged model sees only trusted input and holds tools; a quarantined model reads untrusted content with no tools; a non-LLM controller passes opaque references.
- **Code-Then-Execute** (CaMeL): the privileged model writes a program; values carry capabilities (provenance and allowed readers); policies are enforced at each tool call so untrusted data cannot change control flow.
- **Context-Minimisation**: remove the user's original prompt or untrusted text from context once it is no longer needed.

10.4 A layered defence for a consumer agent

1. **Provenance on every value.** The harness tags what came from the user, from memory, from the user's own accounts and from the open web. The policy engine checks "may this value flow to this sink?" (CaMeL-lite) before every side-effecting call.
2. **Trifecta breaker in the orchestrator, not the prompt.** Any run that has ingested untrusted content and holds private data cannot call an outbound tool (send, post, pay, share a link containing data) without a user approval that shows the exact payload. OpenAI's "Safe Url" check (ask before conversation data is sent to a third party) is the same idea.
3. **Quarantined readers.** Web pages, emails and documents are read by a tool-less model that returns typed fields (dates, amounts, names, a summary), not free text that flows back into the planner.
4. **Injection probes on tool results** that warn the model or ask the user.
5. **An action-alignment classifier** before each side-effecting call: does this action serve the user's original request? This is what lets Anthropic run autonomous browser actions without per-step approval.
6. **Least privilege per step.** Read scopes first, write scopes on demand; tools exposed per skill rather than all at once.
7. **Memory hygiene.** Facts extracted from untrusted content are marked as such and never become standing rules or approval preferences; only the user's own statements can.
8. **Human approval for the irreversible.** Money, outbound messages, deletes and sharing personal data always need a tap on a card that shows exactly what will happen. Watch approval fatigue: if users approve 99% of cards without reading, the cards are too frequent or too vague.

10.5 Identity, credentials and authorisation

- **MCP authorisation on OAuth 2.1**: Protected Resource Metadata (RFC 9728) mandatory; Client ID Metadata Documents preferred; Dynamic Client Registration deprecated; tokens audience-bound (RFC 8707) and keyed to the issuer (RFC 9207).
- **Token vault, so secrets never enter context.** Auth0 Token Vault (30+ providers; the agent holds only an Auth0 token and exchanges it at call time), Arcade (the runtime performs the action and returns only the result; per-user KMS-encrypted vault; just-in-time authorisation as the intersection of agent and user

scope), or your own KMS-backed vault with injection at the egress proxy. Never put tokens in environment variables inside the sandbox.

- **Asynchronous approval (CIBA + RAR).** The backend requests authorisation with `authorization_details` describing the exact action; the user approves on a trusted device; the backend polls for the token. Token vaults do not do approvals by themselves; you need both.
- **Cards and passwords** go through a secure form straight to the vault; they are never typed into chat or shown to the model. Card data is tokenised through a PCI-compliant provider.
- **Enterprise direction:** Okta Cross App Access (ID-JAG) lets an identity provider decide which agent may reach which resource. Low relevance for consumers today, but it is where "agent acting for a signed-in user across apps" is heading.

10.6 Audit and recovery

Every external action is logged with timestamp, target, the evidence (screenshot, confirmation number, merchant email) and the approval that authorised it, viewable by the user. A refund-and-remedy policy for agent mistakes ("if we booked the wrong date, we fix it or refund the fee") is a trust feature competitors will copy; it is cheaper than it sounds if the approval and verification layers work.

11 Evaluation and observability

11.1 Public benchmarks: what is still informative

Benchmark	Tests	Current top (source type)	Status
τ^2-bench	Customer-service tool use with a simulated user	Telecom split ~99% (aggregator); official board ~88%	Telecom saturated
τ^3-Banking	Banking service over a ~700-document knowledge base	Qwen 3.8 Max 55.2%, Opus 5 48.7%, GPT-5.6 Sol 46.9% (official board)	Live frontier
GAIA	General assistant multi-step questions	93.69% vs 92% human (aggregator mirror of HF board)	Saturated
BrowseComp	Hard web research	~90–92% tie (mostly lab-reported)	Near saturation
OSWorld-Verified → OSWorld 2.0	Desktop tasks → long workflows	90.2% → 44.33% binary (official)	Old saturated; new is live
Terminal-Bench 4.0	Agentic terminal tasks	~45% (Fable 5 + Claude Code 44.5%; ranks 1–3 not captured)	Live
SWE-bench Verified	GitHub issue fixing	~80.9% when OpenAI stopped reporting	Deprecated: flawed tests and contamination; ~30% of SWE-bench Pro also reported broken
Humanity's Last Exam	2,500 expert questions	~61% no tools, ~68% with tools	Still discriminative
METR time horizon	Length of software tasks at 50% success	≥16 h (Claude Mythos preview), beyond what the suite measures reliably	Instrument saturated
Vending-Bench 2	Run a simulated business for a year from $500	GPT-6 Astra $15,514.70	Live; documents deception in top scorers
DolphinBench	Memory used to complete tool tasks, with cost and latency	70.67%	New (Mem0-authored)

Takeaways: report **pass^k**, not just pass@1, because consumers need the same task to work every time; prefer privately authored, holistically graded evaluations; and remember that harness choice moves scores by 20+ points.

11.2 Your own evaluation suites

Public benchmarks tell you which models to shortlist. Your own suites decide what ships.

Suite	Contents	Metrics	Gate
Answer	500 items: Indonesian high-school and university maths and physics (UTBK/TKA style), factual lookups, PDF and image questions, Bahasa and code-switched prompts	Exact match; judge score with κ	No regression
Brevity and format	200 simple prompts with reference lengths and expected block types	Length ratio, answer-first score, correct block type	No regression

Suite	Contents	Metrics	Gate
Action	100+ scripted flows on cloned or sandboxed sites (shops, bookings, cancellations, Indonesian marketplaces and billers)	Success pass^3, steps, cost, wrong-action rate, hand-back quality	Wrong-action rate must not rise
Memory	Multi-session scripts: set, change, delete, sensitive, conflicting, cross-context	Task success, false recall, deletion completeness, latency	Deletion must be 100%
Safety and injection	AgentDojo-style pages, phishing forms, poisoned emails, out-of-policy purchases, malicious skills	Utility and attack success tracked together	Zero unsafe actions
Proactivity	Replayed inboxes with known "should notify" and "should not" items	Precision, recall, card action rate	Precision floor
Voice	Real Indonesian recordings with accents, noise, numbers	WER, task success, latency P50/P95	Latency budget

11.3 Methodology

- **Calibrate LLM judges with Cohen's κ** against human labels, not raw agreement; a 21-judge study found exact-match agreement overstates agreement by 33–41 points. Bias-correct judge-estimated pass rates using the judge's measured sensitivity and specificity. A common heuristic: human–human rubric κ of at least 0.6 before calibrating a judge, then judge true-positive and true-negative rates of at least 0.9 per axis.
- **Score trajectories, not just outcomes.** Plan quality, tool choice, arguments, use of outputs, loops. For security, replay full trajectories in an isolated environment and keep the first unsafe action as the failure signal.
- **Run in CI** on every prompt, skill, model or tool change; block merges on regression.
- **Production replay.** Sample real traces (with consent and PII scrubbing), turn every failure into a test case weekly. This is the compounding asset.

11.4 Observability

Tool	Status (Oct 2026)	Recommendation
Langfuse	Acquired by ClickHouse (Jan 2026); remains MIT and self-hostable	Default open-source choice; self-host for PDP data residency
Braintrust	$80M Series B (Feb 2026); eval-first with CI gates	Strong for evaluation workflows
LangSmith	Framework-agnostic tracing, online evals, judge calibration	Good if already on LangChain
Arize Phoenix	OSS on OpenTelemetry/OpenInference; Dynatrace acquisition reported *(unverified)*	Alternative OSS
Helicone	Acquired by Mintlify (Mar 2026), maintenance mode	Avoid for new builds

OpenTelemetry GenAI conventions moved to their own repository (`semantic-conventions-genai`) in June 2026, and no `gen_ai.*` item is marked stable yet. Agent spans exist (`create_agent`, `invoke_agent`, `execute_tool`, memory operations, `gen_ai.evaluation.score`). Instrument with OTel now, pin the convention commit, and treat agent attributes as provisional.

Dashboards that matter: cost per completed task by tier, cache-hit rate, escalation rate, wrong-action rate, approval rate and time-to-approve, hand-back rate, sign-in success rate per site, P50/P95 latency per block type, and tasks completed per weekly active user (the north star).

12 Regulation and compliance

This chapter is a technical summary, not legal advice. Get Indonesian counsel before launch.

12.1 Indonesia

Personal Data Protection Law (UU 27/2022). Fully applicable since October 2024. The implementing regulation, **PP 33/2026**, was issued on 16 Jul 2026 and **takes effect on 16 Jan 2027**; it details supervision, audits, complaints and administrative sanctions. The independent PDP agency has **not yet been formed**: its draft presidential regulation went to the President on 20 May 2026, and experts expect the agency to be effective around 2028. Reports that it has been signed are unverified. Practical obligations for an agent: a lawful basis and consent records for each processing purpose, a data-subject access, correction and deletion API, breach notification within 3×24 hours, data-protection impact assessments for high-risk processing (large-scale profiling and automated decisions qualify), and a data protection officer as you scale. Cross-border transfers need adequate safeguards, which matters when routing Indonesian users' data to US or Chinese model APIs.

AI regulation. Two presidential regulations, a National AI Roadmap 2026–2029 and an AI Ethics regulation, finished inter-ministerial work in May 2026 and went to the State Secretariat in June; there is no evidence either is signed as of October 2026. The ethics regulation reportedly carries no sanctions. The first planned derivative rule is **mandatory labelling or watermarking of AI-generated content**. Seven ministries also agreed guidelines on AI in education, relevant to an exam-help feature.

Children: PP 17/2025 ("PP Tunas") and Permenkomdigi 9/2026. Enforced since 28 Mar 2026. Under-16s cannot hold accounts on high-risk platforms; enforcement started with YouTube, TikTok, Facebook, Instagram, Threads, X, Bigo Live and Roblox. Platforms must provide age-based classification, **reliable age verification (a self-declared checkbox is not enough)**, parental controls and child-friendly privacy defaults; **profiling children's data for advertising is banned**. Whether an AI assistant is classed as high-risk is not established, but every platform must run the self-assessment. For a product whose beachhead includes students aged 15–18, this is a design input from day one: age assurance at sign-up, a restricted mode for minors (no purchases, no companion-like features, stricter memory defaults), and parental consent where required.

Payments. Foreign payment providers need a PJP licence (PBI 10/2025); integrate through a licensed PSP. QRIS rules and fees are set by Bank Indonesia.

Telecoms. +62 numbers require KYC; Indonesian entity and NPWP for business numbers.

12.2 European Union

The AI Act's prohibited practices and AI-literacy duties have applied since 2 Feb 2025, general-purpose model obligations since 2 Aug 2025, and **Article 50 transparency since 2 Aug 2026** (tell users they are talking to AI; mark synthetic content in machine-readable form, with a grace period to 2 Dec 2026 for systems already on the market). The **Digital Omnibus (Regulation 2026/1744)**, in force since 27 Jul 2026, delays Annex III high-risk rules to 2 Dec 2027 and Annex I to 2 Aug 2028. A general personal agent is mostly an Article 50 matter unless used in a high-risk domain (credit, hiring, education scoring). Model-provider duties fall on the labs, not on app builders. GDPR changes from a separate omnibus are still in negotiation.

12.3 United States

California **SB 243** (in force since 1 Jan 2026) regulates companion chatbots: disclose AI where a user could be misled, publish a suicide and self-harm protocol with crisis referral, and for known minors show a break reminder every three hours and block sexual content; private right of action; productivity and customer-service bots are excluded. New York's AI companion law (since 5 Nov 2025) requires self-harm detection and a

not-human notice at session start and every three hours. Colorado repealed its AI Act and replaced it with SB 26-189 (effective 1 Jan 2027), plus a Chatbot Safety Act. Texas TRAIGA has applied since 1 Jan 2026. Executive Order 14365 seeks to pre-empt state AI laws, but no federal statute has passed.

12.4 Compliance switches to build into the product

Switch	Driven by	Implementation
AI disclosure	EU Art. 50, SB 243, NY Art. 47	Clear "AI" labelling in every channel; machine-readable marks on generated media
Consent ledger	PDP Law, PP 33/2026	Per-purpose consent records with timestamps and versions
Access, export, deletion	PDP Law	Self-serve "what you know about me", export, verified deletion across memory, logs, vector indexes and backups
Age assurance and minor mode	PP Tunas, SB 243	Age check at sign-up; restricted mode under 18; parental consent flow; no ad profiling
Crisis protocol	SB 243, NY, duty of care	Detect self-harm signals; respond with care and local crisis resources (verify current Indonesian hotlines before launch)
Data residency	PDP cross-border rules	Regional model endpoints (Singapore), self-hosted open models for sensitive extraction, Indonesian or Singapore data stores
Breach response	PDP Law	72-hour notification runbook, tested
AI content labelling	Planned Indonesian derivative rule	Label generated images, audio and documents now

13 Unit economics

13.1 Assumptions

All figures are my estimates from October 2026 list prices in this document, rounded, excluding taxes and enterprise discounts. A "turn" on the small tier is 20K context tokens (90% cached) and 500 output tokens on Claude Haiku 5.5 or GPT-6 Luna (≈ $0.0006). A Tier-2 task is five workhorse turns of 30K context (90% cached) and 1K output on GPT-6.1 Sol or Sonnet 5.5 (≈ $0.094). A Tier-3 errand is 30 small-model browser steps (40K context, 80% cached, 300 output) plus 10 workhorse planning turns (≈ $0.23). An Opus 5.5 escalation is 40 steps of 60K context with 2K output (≈ $3.00). Rupiah conversions assume about Rp16,500 per US dollar *(assumption)*.

13.2 Cost per active user per month

Item	Light (free tier)	Typical (paid)	Heavy (premium)
Small-tier turns (chat, routing, extraction)	100 → $0.06	150 → $0.09	300 → $0.19
Tier-2 tasks (research, analysis, drafting)	10 → $0.94	30 → $2.81	60 → $5.61
Tier-3 errands (browser actions)	1 → $0.23	6 → $1.35	15 → $3.38
Frontier escalations	0	0	2 → $6.00
Browser hours	0.3 h → $0.02	2 h → $0.10	8 h → $0.80
Residential proxy bandwidth	20 MB → $0.15	100 MB → $0.75	400 MB → $3.00
Sandbox (1 vCPU, 2 GiB)	1 h → $0.08	5 h → $0.41	15 h → $1.24
Voice minutes (Gemini 3.8 Live + platform)	0	20 → $0.50	120 → $3.00
Memory extraction, consolidation, embeddings	$0.02	$0.07	$0.15
WhatsApp utility templates ($0.025 each)	0	10 → $0.25	30 → $0.75
Storage, database, queue, observability	$0.10	$0.25	$0.50
Total	≈ **$1.60**	≈ **$6.60**	≈ **$24.60**

13.3 What this implies

1. **Tier-2 work, proxies and voice now dominate**, not small-model chat. Moving half of Tier-2 turns from Sol to Gemini 3.8 Flash (≈ $0.008 per turn at its introductory price) or to Haiku 5.5 with higher effort would cut the typical user's inference bill by roughly a third. This is what a learned router trained on your own traces is for.
2. **One frontier escalation costs as much as a month of light usage.** Escalate only on evidence (a failed step, a loop, a high-stakes action), never by default.
3. **Proxy bandwidth is the hidden line item.** Block images and video in browser sessions where the task allows; prefer APIs and cooperative endpoints; keep sessions warm instead of re-loading.
4. **Pricing for Indonesia.** A free tier capped on Tier-2 and Tier-3 work (about $1.60 per active user is affordable at a realistic free-to-paid conversion), a paid plan around Rp99,000–149,000 per month (about $6–9), and a premium plan around Rp299,000 (about $18) with heavy-user fair-use caps. At the typical profile, the mid plan has thin gross margin until routing and caching improve; weekly passes (Rp25,000–35,000) suit students around exam periods. Pay through QRIS, GoPay, OVO and DANA via a

licensed PSP.

5. **Fixed costs to budget.** Google CASA assessment (about $500–5,400 per year once you read Gmail server-side), a penetration test before launch, legal review for PDP and PP Tunas, and +62 telephony KYC if you add calling.
6. **Unit-economics guardrail.** Keep inference plus infrastructure under 40% of average revenue per paying user; track cost per completed task, cache-hit rate and escalation rate weekly.

14 The reference stack: a bill of materials

The table gives one primary pick per layer for a small team building a consumer agent for Indonesia and Southeast Asia, an alternate, and what changes at consumer scale (hundreds of thousands to millions of users).

Layer	MVP pick	Alternate	At consumer scale
Mobile client	Expo (RN 0.8x) + `expo/fetch` streaming + native typed-block renderer	Flutter + A2UI renderer	Native modules for voice, on-device models, widgets
Web client	Next.js + Vercel AI SDK	Remix / SvelteKit	Edge rendering, regional deployment
Chat channel	Telegram bot (Rich Messages, draft streaming)	—	Add RCS for verified Android notifications
Notifications	Expo push + WhatsApp utility templates via a BSP	FCM/APNs direct	Per-user interrupt budgets, A/B tested
Voice	LiveKit Agents + Gemini 3.8 Live	Pipecat; cascaded Soniox → LLM → Cartesia/ElevenLabs	Regional media servers (Singapore), self-hosted STT for cost
External surface	MCP server + MCP Apps (runs inside ChatGPT, Claude, Copilot)	—	Listed in app/plugin directories
API and streaming	TypeScript (Node) on Cloud Run or Fly.io (Singapore / Jakarta regions)	Python FastAPI	Kubernetes or Cloudflare Workers + Durable Objects
Harness	Own loop on Vercel AI SDK 7 or Mastra; skills as SKILL.md in git	OpenAI Agents SDK; Claude Agent SDK	Learned router; skill A/B platform
Hosted harness (fallback)	OpenAI Agents API or Claude Managed Agents for long jobs	—	Keep as overflow executor
Model gateway	LiteLLM self-hosted with per-user budgets	OpenRouter for the long tail	Provisioned throughput; self-hosted open models on SGLang
Models	Haiku 5.5 / GPT-6 Luna (T1); GPT-6.1 Sol / Sonnet 5.5 (T2); Opus 5.5 (T3)	Gemini 3.8 Flash; DeepSeek V4.1-Flash; Qwen3.8-27B self-hosted	Fine-tuned small router and extractor on own traces
On-device	Apple Foundation Models; Gemma 4 E2B via LiteRT-LM	Gemini Nano 4 where available	PII scrubbing and intent on device by default
Durable execution	Inngest or DBOS (Postgres)	Temporal; Restate	Actor per user (Durable Objects or Restate)
Sandbox	E2B with auto-pause	Vercel Sandbox; Modal	Self-hosted E2B (Apache-2.0) or Firecracker fleet
Browser	Managed provider chosen by POC on Indonesian sites (Browser Use Cloud, Browserbase or Kernel)	Steel self-hosted	Own Chromium fleet with persistent profiles, residential egress, Web Bot Auth
Desktop companion	Playwright MCP `--extension` / Chrome DevTools MCP	—	Signed desktop app

Layer	MVP pick	Alternate	At consumer scale
Connectors	Google OAuth (Calendar, `gmail.send` first) + Nango or Arcade	Composio; Pipedream Connect	Own verified OAuth clients; CASA renewed yearly
Payments (subscriptions)	Xendit or Midtrans (QRIS, e-wallets, VA) + Stripe for cards abroad	DOKU	Multi-PSP routing
Payments (agent purchases)	Scoped single-use credentials (Stripe Link / SPT) + DOKU MCP for QRIS/VA	Visa/Mastercard agent tokens when live in Indonesia	AP2-style mandate log; issuer partnerships
Memory	Postgres + pgvector + tsvector, own extraction and nightly consolidation	Mem0 or Zep/Graphiti	turbopuffer namespaces per user for mail and documents
Embeddings and rerank	Qwen3-Embedding (self-hosted) or Voyage 4; Voyage rerank-3-lite	Gemini Embedding; Cohere Rerank 4	Self-hosted embedding and reranking
Database	Supabase or Neon Postgres	Cloud SQL	Regional primaries, read replicas
Object storage	GCS or S3 (Singapore / Jakarta region)	Cloudflare R2	Lifecycle rules for transcripts and screenshots
Auth (users)	Supabase Auth / Clerk with Google sign-in and phone OTP	Auth0	Age assurance provider
Secrets and tokens	KMS-backed vault with egress injection	Auth0 Token Vault; Arcade	HSM-backed keys; per-user key isolation
Trust layer	Policy engine + provenance tags + action classifier + approval cards	—	Dedicated red team; injection probes trained on own attacks
Observability	OpenTelemetry → Langfuse (self-hosted)	Braintrust	ClickHouse-scale analytics
Evaluation	Own suites in CI (answer, brevity, action, memory, safety, proactivity, voice)	Braintrust experiments	Production replay pipeline, weekly

15 Build roadmap

The 8 October paper planned 16 weeks for three to four people. With hosted harnesses, MCP-native connectors and cheaper small models, the same scope fits in about 12 weeks if you rent the loop for the first prototype and replace it with your own once evaluation shows where you need control.

Weeks	Milestone	Exit criteria
1–2	Typed-block web chat; own agent loop on AI SDK 7 or Mastra; LiteLLM gateway; search, fetch, sandbox tools; eval harness in CI from day one	70% on answer suite; brevity suite passing
3–4	Router (tier, intent, risk); skills loader with 10 first-party skills; cache-stable prompt layout; Expo app shell	85% answer suite; cache-hit rate above 70%
5–6	Memory v1 (profile, topics, daily log, nightly consolidation); "what you know about me" view with edit and delete; Google Calendar + `gmail.send`	Memory suite 80%; deletion 100%
7–8	Browser agent v1 on a managed provider with persistent profiles; secure forms for codes; vault with egress injection; verifier; Telegram bot	Action suite 70% pass^3; zero unsafe actions
9–10	Approval cards and order cards; trust layer (provenance, trifecta breaker, action classifier); durable schedules and watchers; push + WhatsApp utility templates	First proactive cards live; card action rate tracked
11–12	Payments: subscriptions via QRIS/e-wallets; agent purchases with scoped credentials; minor mode and age assurance; PDP consent ledger; pen-test; cost dashboards	Paid beta; week-4 retention $\geq$30% for activated users
13–16	Voice (Gemini 3.8 Live via LiveKit); MCP server + MCP Apps listing; CASA assessment for Gmail read; learned router v1 from traces	Voice latency P50 $<$ 500 ms from Jakarta; inference cost per task down 30%

Team. A founder owning product, prompts, skills and evaluation; a full-stack engineer (clients and API); an infrastructure engineer (execution plane, security, durable execution); a part-time operations lead writing skills and handling support. A student founder under 18 should also plan who holds contracts, payment-provider accounts and Google verification (a parent or a registered entity), because most providers require an adult or a company.

15.1 Verticals to launch with

Pick verticals where the action ladder is short (APIs or stable pages), the stakes are moderate, and global agents are weak:

1. **Exams and admissions:** UTBK/TKA practice with sandbox-verified maths, university and scholarship deadline tracking from mail and calendar.
2. **Bills and subscriptions:** PLN tokens, BPJS reminders, subscription audit and cancellation in rupiah.
3. **Shopping comparison:** Tokopedia and Shopee price comparison and cart preparation, with the user paying in their own wallet.
4. **Travel:** Traveloka, tiket.com and KAI search with order cards.
5. **Paperwork guidance:** SIM, STNK, passport and BPJS steps with document checklists.

16 Open problems and research directions

1. **Long-horizon reliability.** Binary completion on OSWorld 2.0 is under 50% and zero-intervention success on 8–16 hour tasks was about 35% in mid-2026 *(unverified)*. Research directions: hierarchical checkpoints with automatic verification, learned recovery policies from production failures, and calibrated "I'm stuck" hand-backs.
2. **Learned routing and budgeting.** Generic learned routers underperform hand-designed ones; per-step, multi-turn routing trained on your own traces with cost-success labels (the cheapest configuration that passes evaluation) is open territory. Treat (model, effort, tool budget) as a joint action.
3. **Provable safety at usable utility.** CaMeL shows provable injection resistance at 77% vs 84% task utility. Closing that gap, and extending capability tracking to browsers and memory writes, is the most valuable security research for consumer agents.
4. **Memory evaluation that predicts satisfaction.** Benchmarks are saturated or broken; action-based suites (DolphinBench) are a start. Needed: evaluations of sensitive-fact handling, life-change detection, cross-context preference transfer and deletion completeness.
5. **Self-improving skills.** Mine successful trajectories into skill updates with human review; detect skill drift when sites change; A/B test skills safely.
6. **Agent identity and accountability.** Signed agents (Web Bot Auth), Know-Your-Agent work between Visa, Mastercard and Ant, AP2 mandates. Open questions: revocation, per-task spending identities, and how liability splits between user, agent operator and merchant.
7. **Low-resource language voice.** Independent Bahasa Indonesia, Javanese and Sundanese speech benchmarks do not exist. Building one (with code-switching and noisy mobile audio) would be both a research contribution and a moat.
8. **On-device personal context.** Processing mail and messages on the phone and sending only derived facts could reduce regulatory exposure (including CASA, if Google confirms) and improve privacy; the engineering for mid-range Android phones is unsolved.
9. **Verification agents.** Independent checking of claimed outcomes (confirmation emails, account pages) before telling the user "done", as a separate model with a separate context.
10. **Proactivity that people keep.** Interrupt budgets learned per user; measuring regret ("I wish it had told me") as well as annoyance.

17 Conclusion

The question in October 2026 is not whether an AI can use a computer: the cheapest models now do it, and the labs host the loop. The question is whether people will let one act for them. That depends on three things the model does not provide: a harness that finishes errands reliably and says honestly when it cannot; a trust layer that makes injected instructions, wrong purchases and leaked data structurally hard; and access to the services, payment rails, language and channels of the people it serves. For Indonesia, the opening is real: no global agent integrates QRIS, local wallets, marketplaces or public services, and none is priced for the market. The channel plan must change (own app and Telegram, not a WhatsApp chatbot), the compliance clock is set (PP 33/2026 on 16 Jan 2027; PP Tunas already in force), and the threat is concrete (Meta Muse arriving inside WhatsApp). Build the harness, measure it relentlessly, and win on the parts that are local.

A Key sources

A selection of the primary and most-used sources behind each chapter, all seen during research on 12 October 2026. Aggregator and secondary sources are marked. The full working notes hold several hundred more citations.

A.1 Models and inference

- OpenAI API pricing and models: https://developers.openai.com/api/docs/pricing, https://developers.openai.com/api/docs/models
- OpenAI prompt caching guide: https://developers.openai.com/api/docs/guides/prompt-caching
- Anthropic pricing and models: https://docs.claude.com/en/docs/about-claude/pricing, https://docs.claude.com/en/docs/about-claude/models/overview
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A.2 Harness and protocols

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- A2UI: https://a2ui.org/; AG-UI 1.0: https://www.copilotkit.ai/blog/ag-ui-1.0
- Anthropic, effective context engineering: https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents
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- Anthropic programmatic tool calling: https://platform.claude.com/docs/en/agents-and-tools/tool-use/programmatic-tool-calling
- Google Research, scaling agent systems: https://research.google/blog/towards-a-science-of-scaling-agent-systems-when-and-why-agent-systems-work
- Cognition, multi-agents that work: https://cognition.com/blog/multi-agents-working
- Temporal + OpenAI Agents SDK: https://temporal.io/blog/announcing-openai-agents-sdk-integration

A.3 Execution plane and commerce

- OSWorld 2.0 paper: https://arxiv.org/abs/2606.29537; leaderboard summary *(secondary)*: https://presenc.ai/research/osworld-computer-use-leaderboard-2026
- Gemini computer use docs: https://ai.google.dev/gemini-api/docs/generate-content/computer-use
- Claude in Chrome GA: https://claude.com/blog/claude-in-chrome-generally-available
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A.4 Memory and connected context

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A.5 Interfaces

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A.6 Trust, evaluation and regulation

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Prices, versions and benchmark figures are as reported by these sources on or before 12 October 2026. Verify before committing money or making public claims.